

Superdiffusive dynamics and nested oscillations in visual search

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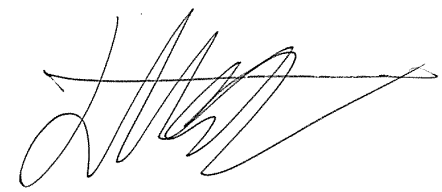
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Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

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Abstract

Gaze dynamics during visual search exhibit rich spatiotemporal structure, providing a valuable window into the neural computations underlying attention and perception. Active vision is an adaptive process in which information gathering is continuously guided by previous observations, engaging distributed neural circuits across the brain. However, existing models of gaze have largely remained divided between biological and predictive descriptions, and between the large-scale dynamics of saccades and the fine-scale movements occurring during fixations. In this thesis, we unify these perspectives within a single physics-inspired model, treating gaze as both a complex dynamical system and an adaptive process of probabilistic computation. Building on the theory of fractional neural sampling (FNS), we understand gaze trajectories through their dynamical signatures which are here demonstrated: global superdiffusion across spatial scales and the nested oscillations of regular saccades which modulate fine spatial sampling within fixations. This method presents a potential basis for non-invasive characterisation of neural function, with possible applications to disorders of attention and perception, including ADHD, schizophrenia, and autism. Broadly, this framework presents a unified account of the rich spatiotemporal signatures of gaze dynamics, inspiring a mechanistic account of neural computation with experimentally testable predictions.

Acknowledgements

I would like to thank my supervisor, Prof. Pulin Gong, for offering such an engrossing research project, balancing exciting results with an emphasis on the *so-what* of it all, and providing guidance across every area of research in my Honours year and in my undergraduate studies. I would also like to thank my co-supervisor, Dr. Brendan Harris, for the endless help and guidance in the most important areas of research, and for introducing me to the Julia hype. I am thankful of my broader research group, for bringing the Madsen Honours office closer to Physics. Among the Physics Honours cohort, I'm particularly glad to be a part of a smaller number who almost always find themselves at the SiFA research office, or else on the short walk to Ralph's café. Finally, and most importantly, I cannot thank enough my family and friends, across Australia and Germany, who—no matter how hard I've made it—always give their unceasing support at any distance.

Statement of student contribution

The central idea of this project was devised by my supervisor, Prof. Pulin Gong, and myself. The theoretical framework of fractional neural sampling (FNS) was previously developed by my supervisor to characterise the rich spatiotemporal dynamics of cortical neural activity. Building upon this theoretical foundation, my supervisor proposed extending the FNS framework to model gaze dynamics. My supervisor also provided a deep understanding of the literature involved—spanning the fields of physics, neuroscience, and machine learning—as well as providing the collaboration with Dr. Pascal Fries, Tim Näher, and Yufeng Zhang, who were invaluable in providing the necessary data and experimental framework to evaluate this work. My co-supervisor Dr. Brendan Harris provided a rich understanding of FNS theory as concerns cortical activity. My co-supervisor also provided practical advice as to computational implementation and tools to be used. My co-supervior provided the basis for the code which structures the numerical solving of the FNS equation in the Julia programming language.

My original contributions to this work are as follows. Broadly, my conceptual additions to this project were in the addition of an informative kernel to couple information gathering to model dynamics, and the interpretation of a framework to span spatiotemporal scales of gaze. Additionally, considering the explored dynamical landscape as a dynamical Bayesian belief distribution was also introduced by myself. The rediscovery of OMT in the eye-tracking data and the model was initiated by myself. Specifically, I provided the careful analysis of kernel-driven rhythmicity, local harmonic oscillations, and Lévy superdiffusion in this system (Chapter 2). The truncated solution of the fractional Fokker-Planck equation in this system, and dynamical approximations to find the theta- and gamma-timescales was done by myself. The writing of the scripts to simulate and analyse the modelled system in the Julia programming language was also done by myself, in the `VisualFNS.jl` project. I devised the methodology to compare the model with eye-tracking data (Chapter 3). This includes the q–order fractional moments compared between the model and the data, which I interpreted as multiscale superdiffusion, through research in literature on SAND. This includes the saccade detection from modelled solutions, as presented and documented in the `VisualFNS_Analysis.jl` project. The construction and methodology of constructing informational metrics was done by myself (Chapter 4). I defined the quantification of efficiency, under and guidance of my supervisors, and provided the interpretation of each considered dynamical signature and compared each to generalised forms of probabilistic computation. The code for defining and calculating these metrics can be found in the `VisualData_Analysis.jl` project.

Generative AI was not used to construct original code, develop theory, or write up and analyse results. Generative AI was used to debug code and in the initial stages of literature search.

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Introduction

The brain is among the most complex systems in science, able to efficiently build a rich internal model of the world from a large quantity of perceptual data. Nowhere is this clearer than in vision. In humans, an estimated one in five cortical neurons is devoted to visual processing and control [1, 2], yet despite this computational power, the efficiency with which information is processed is extreme: for every 1 MB of information received by the retina, only about 40 bit are consciously processed [3–5]. The visual system is therefore highly selective, building a robust and useful representation of the visible world through the high-acuity perception of the fovea, which lies at the centre of visual attention. In maximising the information gathered, the eyes must actively sample the visual environment through dynamics of visual attention, and thus the eyes. The rapid ballistic shifts in visual attention are called saccades, and they organise fixations during which the brain rapidly gathers visual information, before performing the next saccade. Together, they form a sequence of decisions about where to look next, made several times every second during searching to most efficiently build an internal model of a visual scene [6, 7]. Each decision depends on what has already been seen, what is in view, and the goals of the observer. This process of decision-making informing further visual perception is called *active vision*.

This makes gaze a valuable observable for understanding the brain. Gaze provides a direct, low-dimensional, continuous, and non-invasive proxy for neural attention [8–16], which is often difficult to track otherwise. Visual search, in which an observer scans a scene for a target with a determined goal, is among the most demanding and informative settings for studying decision-making in the brain. The subject must balance an exploration–exploitation trade-off, deciding when to continue sampling promising regions and when to discover new ones, while avoiding wasteful re-exploration of depleted regions. This must be done efficiently and robustly within the brain, as the cost of neural computation is high and the brain must be trained to perform well across many tasks [17–21]. The resulting trajectories which are found to be most efficient are rich and structured, with characteristic spatial statistics [22–24], temporal correlations [25, 26], and preferred timescales [8, 9, 27]. Understanding the mechanisms behind them offers a window not only onto vision, but onto the general strategies the brain uses to explore, sample, and decide under uncertainty [28–34].

The rich structure of gaze trajectories now invites a suitably complex physical description. A challenge in approaching this problem has been in combining the near-optimal information-gathering behaviour of gaze with the variability in the particular trajectories taken. In adressing the information-gathering capacity of gaze there has been success in mapping paths which maximise information gathered [35–38], and in addressing the variability of gaze, there has been much success in modelling gaze trajectories through a stochastic process [7, 24]. In this thesis we provide a well-motivated theory in the validity of such a stochastic model which maps complex adaptive cortical activity as described by the theory of fractional neural sampling (FNS) [11, 33] onto the dynamics of gaze. Considering these dynamics abstractly in exploring an informative landscape, we capture the information gathering which drives search tasks generally. By constructing such a model, we are able to ask and propose answers to quantitative questions about gaze dynamics: what are the functional roles of dynamical patterns such as non-Gaussian heavy-tailed fluctuations, adaptive local search strategies, and regular timescales in improving performance during a visual search?

Existing work has approached gaze dynamics largely one scale at a time, both in terms of space and time. Here we present a model which can address gaze dynamics across all scales. Studies of large-scale saccades characterise their timing, amplitudes, and kinematics [6, 39–41]. At the scale of whole search trajectories, stochastic models describe global strategies with non-Gaussian heavy-tailed exploration [42–44], or else through global information-maximising strategies [35, 45–53]. The faster- and shorter spatiotemporal scales of fixational eye movements are then treated as a separate subject, through stochastic descriptions of their high-frequency noise statistics [7, 54, 55] or through their perceptual

role in rapid information-gathering [39, 56], as well as the low-amplitude drifts during fixation [41, 56–58]. The various timescales of attention more broadly, such as theta rhythmicity, form a further literature rooted in neural recordings and behaviour [8, 9, 59]. Each account is valuable on its own terms, but they are developed with different formalisms, different data, and often different assumptions about what the eye is doing.

Each of these approaches leaves a gap, both between biological realistic accounts and empirical accuracy, and between the fine and coarse scales of perception which should each be captured under a single theory. Saccades and fixations are not independent processes: the regularity, size, and duration of saccades are influenced strongly by information gathered during fixation [6, 22, 60]. Models built for a single scale cannot capture this coupling, and their parameters are fitted separately and carry no common physical interpretation. As such, one cannot simply combine such theories into a patchwork theory, as this will not accurately describe the neural control of gaze dynamics. The combined large- and small-scale movements are also rarely measured together, so there are few quantitative constraints on how a unified model should behave across scales [61]. As a result, we lack a single, biologically motivated model in which the heavy-tailed spatial statistics of search, the rhythmic timing of saccades, memory of explored regions, and the structured dynamics of fixation arise from the same underlying biologically-informed mechanism. As such, a unified mechanistic account of search dynamics, going beyond fixation prediction, remains largely unexplored. Such an account must reconcile the phenomena above while remaining biologically plausible, parsimonious, and capable of generating testable predictions about novel or unexplained aspects of visual search.

In this thesis, we begin in Chapter 1 by surveying the relevant literature in studying the rich dynamics of gaze. This includes introducing the broad patterns of gaze trajectories which identify eye movements: in particular their division into jump-like saccades and periods of information gathering called fixations. We then consider the study of diffusive stochastic models which inform this work, followed by the motivating theory of probabilistic computation and neural sampling as driving the information-gathering dynamics of gaze. We then present the model studied throughout this thesis in Chapter 2. We discuss the key dynamical properties present in the model which are found across a large range of spatiotemporal scales: two temporal scales of nested oscillation, and a global spatial superdiffusion. Chapter 3 compares these predictions against real human eye-tracking data, finding validation in the significant regularity of saccade timings, superdiffusion across all spatial scales, and the novel interpretation of a high frequency ocular tremor found in human gaze. In Chapter 4 we present the strength of this model as it provides an understanding of how these dynamical signatures provide insight into patterns of brain function through the map between visual sampling and cortical sampling. These same functional advantages also apply to general probabilistic sampling strategies, such that the efficiency found in visual search can be understood broadly in terms of both a complex dynamical system, and a probabilistic computational system.

Chapter 1: Survey of the Literature

In this thesis, we aim to unify the rich gaze dynamics found during active visual search under a single physics-inspired model, admitting a deeper understanding of neural computation. There is growing success in turning methods from statistical physics and nonlinear dynamics towards the area of computational neuroscience. In particular, the rich and complex spatiotemporal activity of neural circuits inspires similar analysis to unify the dynamics of gaze trajectories across spatiotemporal scales, and present a physics-inspired dynamical understanding of neural computation in search tasks.

In Section 1.1, we review the background for developing a scientific understanding of the dynamics of visual gaze; the data we present in this thesis, the unique spatiotemporal patterns of such data, and the previous methods used to model complex dynamics found in gaze trajectories are each described. In Section 1.2, we introduce the theoretical framework of stochastic modelling, Lévy motion, and the anomalous superdiffusion found elsewhere in complex dynamical systems. Section 1.3 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts; providing the key motivation for developing a mathematical understanding of a gaze trajectory as a proxy for dynamical attention in the brain.

1.1 Gaze trajectories

While healthy human gaze trajectories are well characterised by a small set of dynamical patterns, the challenge is in understanding their interaction, functional roles, and mechanistic origin. The clearest pattern of gaze trajectories is a division into: i) long jumps in positional attention, called saccades, interspersed with; ii) periods of relative stillness called fixations, during which local information about a visual scene is gathered [6, 44, 62, 63]. Eye-tracking devices—such as the Eyelink 1000 used in collecting data for this thesis as shown in Figure 1.1**d)**—provide the high-precision measurements to analyse and model the gaze trajectories through continuous time, both during saccade and fixation.

Where the dynamics of fixation or gaze have been considered, they are treated as separate subject matter to each other, and rarely treated together in single studies. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [7, 66–68]. There are a number of studies examining a rich physical model of fixational dynamics [7, 54, 69], as well as developing the role such

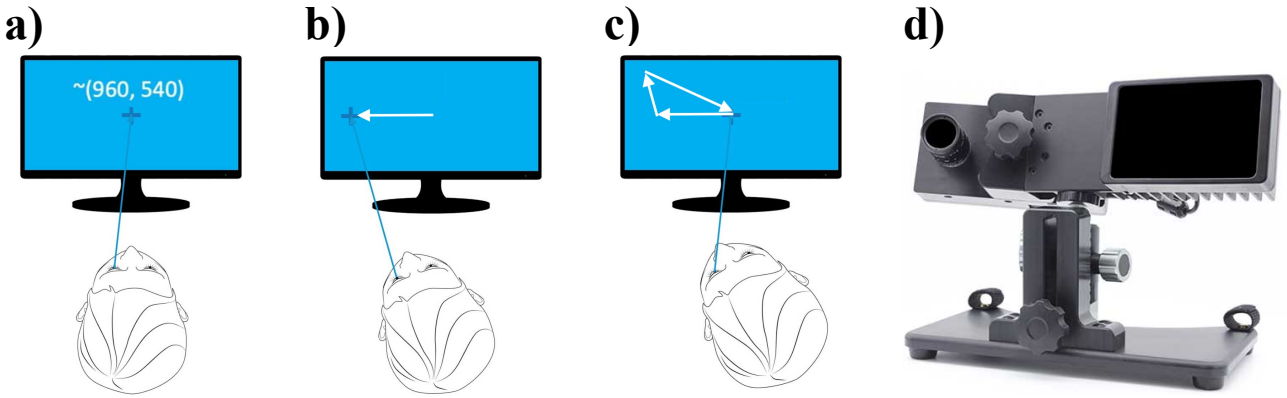


Figure 1.1: A schematic representation of a typical eye-tracking setup. a) shows a subject at the beginning of a typical eye-tracking study, with gaze position starting at the centre of the screen at pixel coordinates $(960, 540)$ px. **b)** shows a saccade from the centre towards the left of the screen as a target is tracked. **c)** shows the return of gaze position to the centre of the screen, as the target returns to the centre. **d)** shows a typical eye-tracking unit, the EyeLink 1000 Plus, as used in the experiment studied in this thesis. The tracker comprises an infrared illuminator and infrared camera ***a)–c)*** *are adapted from [64], while* ***d)*** *is adapted from [65]*

dynamics have in gathering information efficiently [27, 41, 70–73].

1.1.1 Existing models of visual searching dynamics

Progress towards modelling gaze dynamics during visual search spans physics, neuroscience, and machine learning. As such, existing models have been built with very different aims and methods. Cognitive and biologically inspired models explain where attention is drawn through biologically-inspired saliency maps and top-down guidance [74–76]. More normative models ask what an optimally informative searcher would do, selecting saccades by Bayesian belief updating and entropy reduction [37]. We here examine in detail two influential models representing these two model-making paradigms, those being the self-avoiding random walk model of fixational dynamics presented in [7, 77], and DeepGaze III from [48] which utilises a deep learning model to predict saccade positions. A full comparison of a selection of influential and successful models, along with a comparison of their core domains, aims, and key features, is given in Table A.1 in Appendix A.

The self-avoiding random walk model [7] considers the dynamics within a single quadratic potential which confines the gaze into a localised fixation. The gaze position is discretised on a square lattice which is a potential surface. Visited positions increase potential to introduce a self-avoidance condition, which relaxes on a short timescale. Of particular interest to this thesis is the interpretation of the explored potential as a motor map from the neurons in the superior colliculus to motivate a physics-inspired model of gaze mapped from neural activity [78–80]. The self-avoidance condition is mechanistically very similar to the kernel adaptation which we introduce in this thesis to explain rhythmicity in saccades. This work successfully reproduces superdiffusion in fixations, alongside the existence of microsaccades. This model, however, only accounts for fixational motion, and not the long saccades which organise them. Further, although the authors mention tremor, a tremor-frequency peak is not observed.

DeepGaze III [48] models fixation prediction through a state-of-the-art deep convolutional neural network, a map of observed oculomotor biases (such as an observed centre-bias in viewing an image), and a history of previous fixations. This can predict the location of saccades with the highest accuracy to-date across all metrics considered in the standard MIT/Tübingen Saliency Benchmark [48, 81]. This model is highly successful in fixation prediction, and has been used to research patterns in attention through simulation of saccade prediction without requiring human measurement, however it lacks biological or functional usefulness beyond fixation position. The model provides no temporal information, only providing the location of fixations, and does not consider the dynamics of fixations themselves, nor providing any mapping from biologically realistic neural activity onto the dynamics of gaze.

In developing a physics-inspired model of gaze dynamics, the continuous-time dynamics and spatiotemporal patterns of gaze are of great interest. But each of these models leave out those spatiotemporal patterns which have been found in saccade timing, and the high frequency timescale of tremor in fixation. This thesis provides a link between well-studied neural timescales and the dynamics of gaze. It is then valuable to outline the functional interpretations of such frequency bands.

1.1.2 Timescales of neural computation

Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data [82, 83]. Divisions are only approximate, but bands may be thus considered: delta (0.1–3 Hz), theta (4–8 Hz), alpha (8–14 Hz), beta (14–40 Hz), and gamma (40–100 Hz) [82, 83]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to neural computation. Broadly, theta rhythms have been found to modulate and organise bursts in gamma-scale neural activity [9, 33, 84–87]. Theta rhythms have been shown to modulate attention during auditory tasks [14, 88], visual search tasks [8, 9, 89], and when analysing the broad attentional mechanism of the brain [59, 82, 83].

Theta-rhythmic cycles are finding strong interpretation in the cognitive neuroscience literature as

organising and communicating information sampled from the environment for integration into the internal model of the brain [9, 84–87]. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [14, 59], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [8, 9]. Accordingly, the firing patterns which define the lowest level of computation carried out in the brain are regularly found to be gamma in timescale, with broadband signals being found within 40–100 Hz in a wide range of measurements of neural activity during cognitive tasks [90–94]. The coupling between the phase of theta activity and the amplitude of gamma activity is known as theta-gamma coupling (TGC). This phenomenon, a specific form of phase-amplitude coupling (PAC), is schematically represented in Figure 1.2. The evidence and theory behind TGC are described in a systematic review by Ursino and Pirazzini [92]. This thesis provides a computation view of the function of TGC the within the search dynamics of gaze as modulating periods of fixational attention on a slow theta-scale, during which information is dynamically sampled on a gamma scale.

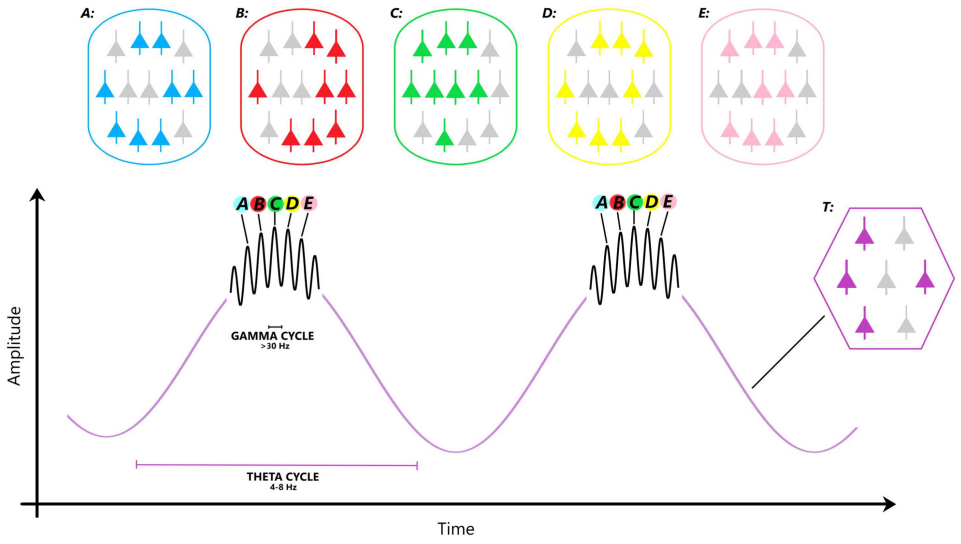


Figure 1.2: A schematic representation of theta-gamma coupling (TGC). Here, neural population patterns $\mathbf{A} - \mathbf{E}$ are activated in sequence on the timescale of a gamma oscillation $f > 30$Hz, triggered by the phase of a modulating theta wave of neuron population $\mathbf{T}$ on the theta timescale of $4 - 8$Hz. Such a formulation of TGC as correctly ordering signals has its foundations in the work of Lisman and Jensen [95], and was subsequently expanded to describe the dynamic neural mechanism during sensory exploration to modulate close attention upon individual centres of focus [96]. *Schematic adapted from Figure. 2 of Ursino and Pirazzini [92]*

1.2 Diffusive processes

In formalising empirical observations of human gaze and neural activity into a mathematical model, this thesis relies on the framework of non-equilibrium statistical mechanics and stochastic calculus. When a system is driven by a massive number of microscopic interactions—such as the complex interconnected firing of a neural population—modelling the exact deterministic dynamics becomes intractable. Instead, statistical physics provides the tools to coarse-grain these high-dimensional systems into low-dimensional stochastic differential equations. This section establishes the mathematical progression necessary to model scale-free visual inference. We first review the classical paradigm of normal diffusion and Brownian motion (Section 1.2.1), which serves as the foundational model for standard diffusive physical systems. We then examine how systems of interest frequently violate the classical assumptions of normal diffusion and Brownian motion, leading to an understanding of the presence and role of anomalous diffusion in many complex dynamical systems (Section 1.2.2). Finally, we introduce Lévy stable motion (Section 1.2.3), a class of superdiffusive fractional stochastic pro-

cesses that mathematically capture the heavy-tailed, intermittent dynamics characteristic of optimal visual search and neural activity.

1.2.1 Brownian motion and normal diffusion

Statistical modelling has been fundamental in understanding complex dynamical systems: separating the dynamics of macroscopic observables into deterministic terms and stochastic terms reduces chaotic high-dimensional systems to a small number of variables of interest [97, 98]. Stochastic processes are often the most natural, tractable, and accurate modelling tools when examining high-dimensional chaotic systems across physics [99–101], machine learning [98, 102, 103], and neuroscience [30, 31, 104]. The universality of these stochastic models is a result of the central limit theorem applied to the fluctuations in these systems: the distribution of the summed effect of a large number of identical and independently distributed (i.i.d) random variables of finite mean and variance will approach a Gaussian [105–107].

The most ubiquitous stochastic process is central to statistical physics: the Brownian motion W_t, equivalently called the Wiener process, which consists of stationary and independent Gaussian increments across all non-overlapping time intervals. A Brownian motion is defined only by its timescale, with a variance over trajectories after time t of t itself, a scaling called normal diffusion. The universality of Brownian motion can be seen in the variety of processes which it can model: from the movement of a macroscopic particle in a non-turbulent fluid [97] to the gravitational influence of an orbiting collection of stars on supermassive black holes [108, 109]. The generalisation of the Brownian motion for heavy-tailed i.i.d variables with diverging variance is a Lévy flight, and is described in detail in Section 1.2.3. The comparison of these processes is shown in Figure 1.3.

Langevin dynamics of Brownian motion

Brownian motion of a variable $\mathbf{x}_t$ of arbitrary dimension through time t about a potential landscape $V(\mathbf{x})$ can be described in terms of the overdamped Langevin equation:

$$d\mathbf{x}_t = -\gamma \nabla V(\mathbf{x}_t)dt + \sqrt{2\gamma}dW_t \tag{1.1}$$

where the constant γ describes noise strength [101, 110–112] and dW_t is a Brownian motion. Although the dynamics of $\mathbf{x}_t$ is described stochastically, the dynamics of the corresponding probability distribution $P(\mathbf{x}, t)$ is deterministic and can be described exactly. Such an equation of motion of $P(\mathbf{x}, t)$ under any stochastic dynamics defining $\mathbf{x}_t$ is called the Fokker-Planck equation [113–118]. In the case of Equation (1.1), the Fokker-Planck equation is [113, 114, 117]

$$\partial_t P(\mathbf{x}, t) = \gamma \nabla \cdot (P(\mathbf{x}, t)\nabla V(\mathbf{x}) + \nabla P(\mathbf{x}, t)) \tag{1.2}$$

whose unique stationary distribution is the Boltzmann-Gibbs distribution

$$\pi(\mathbf{x}) \propto \exp[-V(\mathbf{x})] \tag{1.3}$$

which yields the important relationship, where a Langevin potential can be constructed from a normalised target density via $V(\mathbf{x}) = -\ln \pi(\mathbf{x})$, which we will make use of in this thesis in constructing saliency maps from empirical data where only the likelihood of fixations $\pi(\mathbf{x})$ is given. Importantly, the Langevin system Fokker-Planck Equation (1.2) will describe the diffusion of a sharply peaked initial distribution as it widens, and the statistics of this diffusion from a sharp initial condition define normal diffusion as follows. The solution to the Fokker-Planck equation from an exact initial condition $P(\mathbf{x}, 0) = \delta^n(\mathbf{x} - \mathbf{x}_0)$ is dominated over short times by the ∇P term as $\nabla P \gg P\nabla V$, reducing the Fokker-Planck equation to the heat equation $\partial_t P = \gamma \nabla^2 P$. As such, the leading-order short-time solution is that of such a heat equation

$$P(\mathbf{x}, \Delta t) \propto \exp\left[-\frac{\|\mathbf{x} - \mathbf{x}_0\|^2}{4(\gamma \Delta t)}\right] \implies \mathrm{Var}[\mathbf{x}(\Delta t)] \propto \gamma \Delta t \tag{1.4}$$

which demonstrates the interesting behaviour which defines normal diffusion: the short-time scaling of variance in position will be linear, as $\mathrm{Var}[\mathbf{x}_{\Delta t}] = 2\gamma\Delta t$. Thus the diffusion coefficient γ can be thought of as the timescale of the Brownian motion, as it defines the rate of linear diffusion of variance with time. This linear scaling of the variance of trajectories with time defines normal diffusion [119–122], and understanding those systems whose fluctuations do not reproduce such normal diffusion is central in understanding stochastic motion complex dynamical systems.

1.2.2 Anomalous diffusion

The universality of normal diffusion in systems which are driven by Brownian motion as above has been studied closely, most influentially as in the movement of particles in a fluid [100, 109, 119, 123]. These dynamics inform the definition of normal diffusion through this variance scaling with time, rigorously calculated as the mean squared displacement (MSD) $\langle \|\Delta\mathbf{x}(\Delta t)\|^2 \rangle \propto \Delta t$. However, increasingly interesting physical systems have been found which diffuse nonlinearly in time, a situation known as anomalous diffusion, with an MSD scaling as:

$$\langle \|\Delta\mathbf{x}(\Delta t)\|^2 \rangle \propto \Delta t^{\lambda_2}, \quad \lambda_2 \neq 1 \tag{1.5}$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \lambda_2 < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \lambda_2 < 2$ where diffusion scales faster than normal. The case of $\lambda_2 = 2$ is known as ballistic diffusion, as it is found in the deterministic case of constant velocity $x = vt$.

Systems displaying anomalous diffusion are self-similar, and exhibit scale-free fractal spatiotemporal fluctuations, as a result of their continuous-time power-law diffusive scaling. It has been found to be advantageous in the study of anomalous superdiffusion to capture this diffusive self-similarity across scales through an extension of the λ_2–scaling of MSD to a spectrum of absolute moments $\lambda(q)$. This spectrum describes the diffusive self-similarity from the small spatial scales (low q) to large spatial scales (high q). Such a spectrum of absolute moments $\lambda(q)$ to capture diffusive scaling is [120]:

$$\langle \|\Delta\mathbf{x}(\Delta t)\|^q \rangle \propto \Delta t^{\lambda(q)}, \quad q > 0 \tag{1.6}$$

where q is the order of the fractional absolute moment, with $q = 2$ recovering the MSD, and $q = 1$ yielding a mean absolute displacement (MAD), and so on for any value $q > 0$. Systems exhibiting so-called strong anomalous diffusion (SAND) or multifractality will have a nonlinear spectrum $\lambda(q)$, which is interpreted as different statistical scaling properties on different scales (for instance, a Brownian particle which exhibits rare large jumps [124]). A process which is uniformly self-similar on all scales will be defined by a linear spectrum $\lambda(q) = \frac{q}{\alpha}$. We will see that both our data of gaze dynamics and the model we develop are monoscaling across many scales of q (see Section 1.2.3 and Section 2.4).

Anomalous superdiffusion

Of great importance in studying the dynamics of visual search, the pattern of superdiffusive motion is found in many complex biological systems. We model the efficient exploration of gaze dynamics through Lévy motion, which is the standard dynamical model of superdiffusion in physics and is presented in detail in Section 1.2.3. Such Lévy superdiffusion has a strong link to the problem of designing a strategic search for scarce resources. The optimality of a superdiffusive process in a search has been reproduced in diverse functional search models [38, 125, 126], in a population of diverse searching agents with a range of superdiffusive exponents λ_2 [127] as well as in a large study against several different search conditions [125], and have been reported in the empirical search paths of biological organisms from the flights of drosophila flies [38, 128], albatross' searches [129, 130], and strongly across many marine predators [131, 132]. This thesis provides a biologically realistic and well-motivated superdiffusive search for research into a case for optimal Lévy motion. See Reynolds [133] for an in-depth discussion of a richer view of biological Lévy superdiffusion.

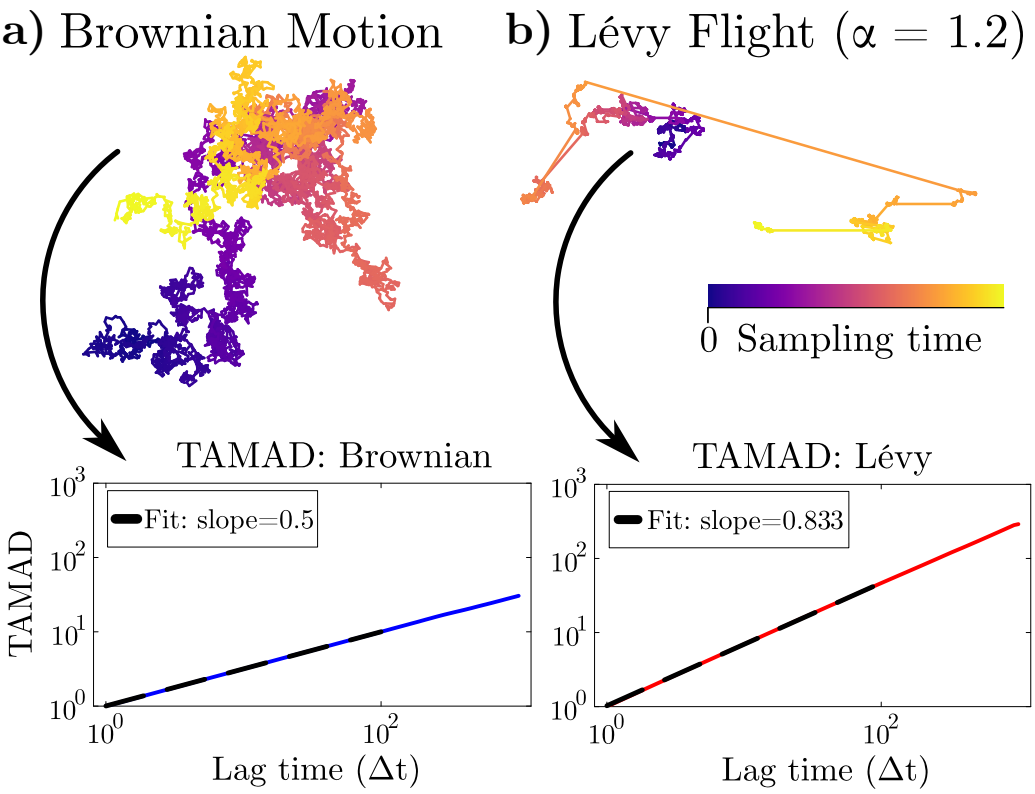


Figure 1.3: Schematic representations of a normally diffusive Brownian motion and a Superdiffusive Lévy flight. a) Schematic representation of a 2D Brownian motion, along with its time-averaged mean absolute displacement (TAMAD), which is the $q = 1$ mean fractional displacement as in Equation (1.6). Normal diffusion is here present as a power law slope of exponent 1/2, giving diffusive exponent $\lambda_1 = 1/2$ and $\alpha = 1/\lambda_1 = 2.0$. **b)** Schematic of a superdiffusive 2D Lévy flight with characteristic exponent of $\alpha = 1.2$, and a MAD slope of $\lambda_1 = 5/6 = 0.833\ldots$ corresponding to the theoretical superdiffusive coefficient $\alpha = 1/\lambda_1 = 1.2$.

1.2.3 Lévy motion

To mathematically capture the intermittent, scale-free nature of visual search, we turn to the Lévy flight: a superdiffusive stochastic process that permits close analysis, due to it's very foundational mathematical definition. Here, the Lévy flight L_t^α is a stochastic process defined through it's increments being drawn from a heavy tailed symmetric α-stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution), where α is called the stable exponent and defines the strength of superdiffusion produced. $\mathcal{S}\alpha\mathcal{S}$ distributions describe the limit of summation of i.i.d infinite-variance variables, as a result of the generalised central limit theorem, making them ideal to describe a universality class of superdiffusive processes. The α value defining the $\mathcal{S}\alpha\mathcal{S}$ distribution quantifies the heaviness of its asymptotic tails, with low values of $\alpha \to 1$ producing stronger superdiffusion, and larger values $\alpha \to 2$ eventually reproducing the normally diffusive Brownian motion. The exact form of the $\mathcal{S}\alpha\mathcal{S}$ distribution of jumps $\Delta\mathbf{x}$ is defined by the characteristic function $\mathbb{E}[e^{i\Delta\mathbf{x}}] = \exp[-(\gamma t)^\alpha \Delta\mathbf{x}^\alpha]$, which will yield asymptotic power law tails with $P(\Delta\mathbf{x}) \sim \|\Delta\mathbf{x}\|^{-(1+\alpha)}$. As a result, the Lévy flight L_t^α after time t is defined by its fractional scaling properties and α–stable statistics

$$L_t \overset{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \tag{1.7}$$

where $\overset{d}{=}$ denotes equal in distribution. In the limit $\alpha \to 2$, the Lévy flight reduces to a scaled Brownian Motion $L_t^{\alpha\to 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges and has mean 0, whereas if $0 < \alpha \leq 1$, then both the variance and mean diverge. In analysing the distribution of a Lévy flight, one must generalise the standard Langevin Fokker-Planck equation (1.2) to a Lévy motion using fractional calculus to capture nonlocal transport. To analyse this nonlocal behaviour of superdiffusion, the

fractional Fokker-Planck equation is used, which yields the nonlocality, and scale-free behaviour of trajectories of Lévy flights (see the appendices to [33]). For an analysis of the fractional Fokker-Planck equation in this thesis, see Section 2.4. Many of the simple analytic properties of the Brownian motion are no longer true of a Lévy flight. Notably, since these flights have infinite variance in steps sizes, the mean squared displacement diverges. Thus their superdiffusion is better measured using the extension of MSD to the spectrum of fractional moments defined in Equation (1.6):

$$\langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle \propto \Delta t^{\lambda(q)}, \quad \lambda(q) = \frac{q}{\alpha} \quad \alpha > q > 0 \tag{1.8}$$

where q is the order of the fractional absolute displacement. Thus, this predicts that an α–Lévy flight is monoscaling with spectrum $\lambda(q) = \frac{q}{\alpha}$ for $q < \alpha$. As such, for Lévy flights with exponents $\alpha > 1$ superdiffusion is well-characterised by the mean absolute displacement, where $q = 1$ in Equation (1.6), and shown in Figure 1.3. We will see that the Lévy dynamics through this thesis in a bounded potential, with momentum-dynamics, and lightly smoothed paths produce monoscaling superdiffusion beyond $q = \alpha$, to all scales q. Importantly in understanding this monoscaling, the Lévy-Itô decomposition theorem describes how a Lévy flight produces superdiffusion through the common large jumps it exhibits. The Lévy-Itô decomposition theorem states that the Lévy flight can be expressed as the sum of two processes: [134]:

1. A compound Poisson process of large jumps with $\|\Delta \mathbf{x}\| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number in any finite time period.
2. A square-integrable jump process with mean 0 with $\|\Delta \mathbf{x}\| < 1$. This process is then finite-variance, and will approach a Brownian motion when isolated from the full continuum of jumps.

This division is useful in analysing the process in the context of nonlocal stochastic exploration in Lévy flights, which is modelled by the compound Poisson process of large jumps $\|\Delta \mathbf{x}\| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $\|\Delta \mathbf{x}\| < 1$. In the context of visual searching, such a division naturally inspires the division between large jumps of attention, and periods of prolonged attention. As such, we are interested in why such superdiffusive dynamics may have a powerful in computational advantage in exploring an abstract unknown landscape, especially in the context of visual searching.

1.3 Probabilistic computation

Computation in the brain must account for high levels of uncertainty. Sensory inputs are noisy, high-dimensional, and ambiguous, so inference in the brain must be represented by some distribution of activity and decisions, rather than a single deterministic decision. The stochastic dynamical exploration of an information landscape, known as probabilistic computation, offers a practical framework to understand the variability of a sampling process, to explore an internal model of the world and inform decision-making. Considering neural activity, the explored landscape represents a target distribution of neural firing, and a sampling process approximates this distribution through stochastic samples (see Figure 1.2). From an algorithmic perspective, an effective sampler should produce near-independent samples which decorrelate quickly and explore robustly and dynamically [31, 34].

This paradigm of a sampling process as probabilistic computation underlies modern machine learning and statistics, where generative models learn patterns from sparse samples of large multivariate distributions [31, 135]. It has also gained traction in for biological computation, in which the variability of neural population activity is not a nuisance but a functional signature of an internal model of perceptual uncertainty [32, 136, 137]. This section outlines the neural sampling hypothesis as an interpretation of variability in neuronal firing patterns (Section 1.3.1), introduces fractional neural sampling (FNS) as a model of the spatiotemporal structure of that variability, and then extends the framework to visual search as a sampling process (Section 1.3.2).

1.3.1 Neural sampling

The neural sampling hypothesis proposes that the variability of cortical responses is a functional feature of computation [30]. Stochastic activity patterns, otherwise regarded as noise, are thus considered to be samples drawn from a posterior distribution over possible sensory interpretations [30, 104]. Uncertainty is then encoded in the spread of these responses across time and trials, with wider response distributions reflecting greater uncertainty [30, 31, 104]. The hypothesis has accounted for network-level patterns of activity in V1 better than alternative theories [30, 104, 138], and energy-based models offer a biologically plausible implementation [34].

Variability need not arise from true stochastic noise. Spatially extended, balanced excitatory–inhibitory (E–I) circuits can generate rich variability intrinsically, producing localised activity patterns that wander and switch between locations [11]. Gaussian and Poisson noise models drive diffusive exploration which is normally diffusive with no dominant timescale. In these models, sampling mixes slowly and are highly correlated, tending to remain trapped in a single mode of a multimodal distribution [139, 140]. The dynamics of E–I networks appear to be richer than this, which has motivated functional models that treat the structure of neural variability as part of the computation itself.

Fractional neural sampling (FNS)

FNS is one such model [33]. It describes the centre of mass of a localised spiking pattern as a Lévy flight (see Section 1.2.3) whose long-time superdiffusive statistics match an uncertain and complex multimodal target probability distribution more quickly than similar models based on Brownian motion. The model is called fractional because the non-local dynamics of this superdiffusive motion require fractional calculus to describe [141] (see Section 2.4 for such a treatment). FNS dynamics capture the rich spatiotemporal dynamics of cortical activity patterns of travelling waves through the E–I circuit models of neural activity, which capture the neural coupling between theta-rhythmic organisation of gamma-scale sampling [11], as shown in Figure 1.2. FNS therefore links the rich temporal and spatially superdiffusive dynamics of neural firing patterns to the powerful neural sampling framework.

The principal advantage of FNS dynamics over standard continuous random walks in describing neural sampling in the brain is its ability to efficiently sample complex, dynamical, and multimodal distributions with far-apart modes, a feature of natural environments replete with multiple distinct salient objects. While Brownian samplers are trapped by high potential energy barriers, the heavy-tailed jumps of FNS allow the sampler to switch freely between modes, balancing exploitation of known high-probability regions with exploration of uncertain ones [11, 142, 143]. Furthermore, because the movement of the eyes is driven by E–I neural circuits that are expected to preserve key features of the model, the FNS description can be applied to gaze dynamics.

1.3.2 Visual sampling

This thesis presents visual search tasks as such a paradigm where a sampling model is well motivated, and driven by the underlying neural sampling process as well. The visual system is evolved to actively and efficiently process noisy information and evidence from the external world, and plan future search paths to gather further noisy evidence [28, 37, 144, 145]. The brain is hypothesised to create a map of the most interesting areas of a visual landscape, to decide where to look first, called a saliency map [3]. Here, saliency is equivalent to the Boltzmann log-likelihood potential presented in Equation (1.3) from the probability distribution of fixations $p(\mathbf{x})$, $S(\mathbf{x}) = -\log(p(\mathbf{x}))$, where such a saliency map informs the exploration of gaze as it samples highly salient areas rapidly [3, 10]. As such, undirected visual searching of an image will show the areas most salient to the brain, allowing insight into the computational sampling mechanism of the visual system. In particular, search tasks require sophisticated integration of the cost and expected information gain of decisions, which invites fruitful decision analysis [146]. Empirical experiments show that eye movements show good agreement with greedy local information maximising fixations, given well-motivated visual sensitivity based on human physiology [28, 37, 147]. Any development of a physics-inspired and biologically-motivated model of visual search must then balance informational efficiency with accurate physical description.

Chapter 2: A Unifying Model

We develop a physical theory of gaze which integrates informational awareness and accurate consideration of the statistics of search dynamics. The chapter is organised as follows: Section 2.1 defines the model based from FNS dynamics before presenting the search task used as the paradigm throughout this work. Section 2.2 introduces the dynamical-informative kernel which is novel to this work, giving its modelling motivations (Section 2.2.1) and the regular information-driven temporal structure of transitions between regions of interest as they are explored through informational kernel deposition (Section 2.2.3). Section 2.3 derives and quantifies the prediction from neural dynamics that localised trajectories produce high-frequency simple harmonic oscillations. Section 2.4 treats the defining characteristic of FNS dynamics, Lévy superdiffusion, deriving the short-time noise-driven scaling of fractional absolute moments and the monoscaling of self-similar superdiffusion which the model presents. Comparison of each of these dynamics with eye-tracking data is deferred to the next chapter.

2.1 Model setup

Mathematical formulation of gaze dynamics

The dynamical model of gaze dynamics, inspired by and developed from the FNS theory of neural dynamics (see Section 1.3.1, [33]), models eye movement trajectories as an overdamped Langevin system driven by an adaptive potential landscape $V_t(\cdot)$, which is updated by an information gathering kernel $k(\cdot)$. The full system governing modelled gaze dynamics throughout this work is

$$\begin{aligned} d\mathbf{x}_t &= -\gamma b_\alpha(\mathbf{x})dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta b_\alpha(\mathbf{x})dt, \\ \partial_t V_t(\mathbf{x}) &= \frac{1}{\tau} k(\mathbf{x} - \mathbf{x}_t), \qquad V_0(\mathbf{x}) = S(\mathbf{x}). \end{aligned} \tag{2.1}$$

Due to the equivalent relationship of a Langevin potential to its targeted distribution in Equation (1.3) and the definition of saliency in visual sampling as in Section 1.3.2, the potential V_t defines a continuous-time updating internal-state belief distribution of the search target:

$$\pi_t(\mathbf{x}) \propto \exp\left[-V_t(\mathbf{x})\right]. \tag{2.2}$$

The terms of Equation (2.1) are as follows. Let $\mathbf{x}_t$ denote the two-dimensional position of visual attention, and $\mathbf{p}_t$ a generalised auxiliary variable analogous to momentum, which encodes short-scale memory effects. The parameters α, β, γ modulate the FNS dynamics: $\beta \geq 0$ is the coupling strength between $\mathbf{p}_t$ and $\mathbf{x}_t$, $\gamma > 0$ is the noise strength parameter as in the standard Langevin system (Equation (1.1)), and $0 < \alpha \leq 2$ is the Lévy tail exponent, which defines the heaviness of the tails from which noise is drawn, and equivalently the strength of superdiffusion as in Sections 1.2.2 to 1.2.3. The driving noise process dL_t^α represents independent symmetric α-stable ($\mathcal{S}\alpha\mathcal{S}$) Lévy fluctuations. The constant c_α corrects the scale of the gradient drift, as derived below. The third line of Equation (2.1) governs the adaptive landscape: τ controls the rate of information updates to the dynamical-informative potential $V_t(\mathbf{x})$, $k(\cdot)$ is the information-gathering kernel centred on the current position $\mathbf{x}_t$, and $V_0(\mathbf{x}) = S(\mathbf{x})$ is the initial landscape. The nature and purpose of the dynamical updates $k(\cdot)$ is described in Section 2.2.

The drift terms in Equation (2.1) follow from the FNS theory. A deterministic drift $b_\alpha(\mathbf{x}_t)$ drives the system, which is itself driven by Lévy noise, toward an invariant target distribution $\pi(\mathbf{x})$ over long times. As derived in [33], the exact non-local drift for arbitrary $0 < \alpha < 2$ is formulated in terms of

the Riesz $(\alpha-2)$–fractional derivative $\mathcal{D}_{\mathbf{x}}^{\alpha-2}$:

$$b_\alpha(\mathbf{x}) = \frac{\mathcal{D}_{\mathbf{x}}^{\alpha-2}[\pi(\mathbf{x})\nabla_{\mathbf{x}} \ln \pi(\mathbf{x})]}{\pi(\mathbf{x})}, \tag{2.3}$$

where these nonlocal fractional dynamics are treated in detail in Section 2.4. However, due to the computationally intense calculation of the nonlocal operator $\mathcal{D}_{\mathbf{x}}^{\alpha-2}$, in this work we adopt the local gradient approximation, which becomes exact as $\alpha \to 2$, as given in Equation (14) of [142]:

$$b_\alpha(\mathbf{x}) = c_\alpha \nabla_{\mathbf{x}} \ln \pi(\mathbf{x}), \qquad c_\alpha = \frac{\Gamma(\alpha-1)}{\Gamma(\frac{\alpha}{2})^2}. \tag{2.4}$$

Substituting the time-dependent belief π_t of Equation (2.2) into Equation (2.4) gives $b_\alpha(\mathbf{x}) \approx -c_\alpha \nabla V_t(\mathbf{x})$, which is the drift appearing in the first two lines of Equation (2.1).

Empirical protocol

To validate this modelled computational framework against human behaviour, we emulate the visual foraging task designed by Näher et al. [61], which requires subjects to sequentially scan an array of dots to locate a hidden target. This is done by choosing biologically accurate values for parameters $\alpha, \beta, \gamma, \tau$ and the information gathering scale of $k(\cdot)$. The task itself is encoded in the initial saliency map $V_0(\mathbf{x}) = \log \pi_0(\mathbf{x})$. These parameters and choice of $\pi_0(\mathbf{x})$ are described in the final part of this section.

In the human-testing protocol examined by Näher et al. [61], trials commenced with a pre-search fixation period of duration drawn uniformly between 2500–3500 ms. Following a 100 ms duration of no stimulus, participants were presented with a search array of white dots, each individual dot subtending 0.3° of visual angle on the screen. Exactly 20% of these dots functioned as true targets, which were visually identical to non-targets, and subjects were allotted a maximum search time of 5000 ms to successfully foveate a target, at which point an audio cue would indicate a success, and the subject would be required to fixate for another post-fixation period of 2500–3500 ms. To investigate the influence of spatial structure, the study compared a REGULAR task—where dots were spaced identically to the centers of an invisible hexagonal tiling—with an IRREGULAR task, which randomly redistributed the same number of dots across the screen to break spatial regularity. Assuming a standard viewing distance of 57.3 cm (where 1° ≈ 1 cm) and a monitor pixel density of 36.22 px cm^{-1}, spatial measurements translate directly between pixels and degrees using a scaling factor of 36.22 px °$^{-1}$.

Data-informed parameter constraint & model setup

The empirical conditions from Näher et al. [61] are translated into numerical model parameters using spatial units of pixels (px) and temporal units of seconds (s). The baseline stimulus potential $S(\mathbf{x})$ is constructed on a 2D grid of Gaussian points of interest, representing quadratic potential wells over a 16×16 square grid (256 wells total), with 5 ($\approx 2\%$) randomly designated as targets. Based on fixational eye movement distributions, the standard deviation of each well is set to $\sigma_{\text{well}} = 3$ px. The average axial distance between wells is fixed at 110 px ($\approx 3.04°$ visual angle), bridging the dense (2.3°) and intermediate (4.5°) spacings in the REGULAR human search condition in Näher et al. [61].

The baseline parameters for standard simulation runs are calibrated as follows:

- $\alpha = 1.2$: Captures the characteristic search superdiffusion observed in Section 3.3 of $\alpha = 1.12$ while maintaining numerical stability.
- $\sigma_k = 80$ px: Defines an adaptive spatial acuity kernel by the Gaussian $k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right)$. This matches the radial spatial scale of human visual acuity of $\approx 2.5°$, corresponding to 87.2 px at a 50 cm viewing distance with 4 px mm^{-1} density. Kernel dynamics are discussed in Section 2.2.2.
- $\tau = 2 \times 10^{-4}$ s: Set to yield theta-rhythmic saccadic transitions, as derived in Equation (2.8)

and evaluated in Section 3.1. These parameters give a theta rhythm of $f_\theta = 7.03\,\text{Hz}$ and $t_\theta = 142.22\,\text{ms}$

- $\beta = 1000\ \text{px}\,\text{s}^{-1}$ and $\gamma = 1000\ \text{px}^{\alpha}\,\text{s}^{-1}$: Recovers stable gamma-rhythmic harmonic oscillations within the oscillatory regime for 3 px wells (Section 3.2). The gamma frequency produced using Equation (2.11) is $f_\gamma = 74.10\,\text{Hz}$

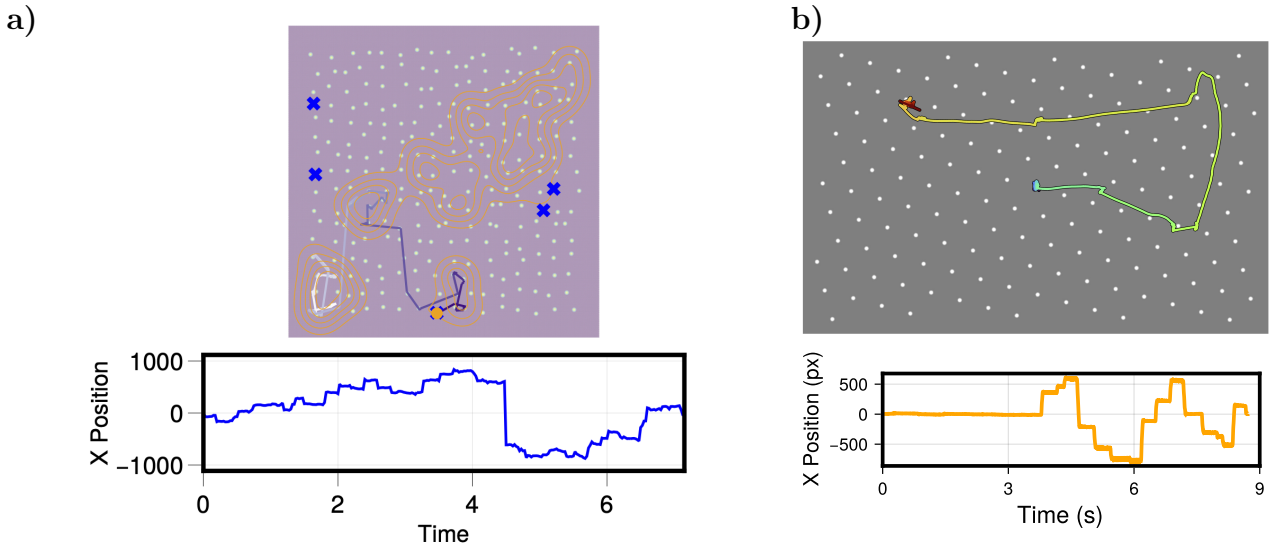


Figure 2.1: A comparison of the modelled search task with an example gaze trajectory from a real human trial. a) The modelled gaze trajectory, with gaze centre marked in orange, and a short history of it's path. All 3 px wide Gaussian points of interest are shown within a heatmap, with peaks in green. Possible targets, the fixation onto which marks a successful trial, are marked by blue crosses. **b)** An example gaze trajectory from the empirical study of 22 human subjects, whose methodological framework is used to study the model throughout this thesis. This study by Näher et al. [61] is described in Section 2.1

2.2 Dynamical-information adaptation

The component of our model of gaze here which connects dynamics to information is the dynamic-informative kernel. It is the principal extension of FNS work towards a full computational model of visual search during active vision [11, 33]. We call the kernel dynamical-informative because it couples two processes: the dynamics of the FNS sampler as it explores a landscape, and the information gained during that exploration when interpreted as active search. Its development therefore draws on two bodies of literature. The first is statistical physics and computational sampling, where an analogous process of kernel deposition is called metadynamics (see Barducci et al. [148] for its introduction and Bussi and Laio [149] for the "well-tempered" variant, which balances sampling accuracy against ergodicity). The second is the psychophysics of how spatial acuity limits information gathering during search, and how information informs the dynamics of search.

The section proceeds as follows: Section 2.2.1 states the problems and motivations which inform the addition of this kernel. Section 2.2.2 derives its form from Bayesian information gathering. We then examine the spatial scale of this information gathering from a consideration of human visual acuity. Section 2.2.3 analyses the resulting dynamics and shows that they encode the theta-rhythmic timescale of saccades studied in Section 3.1, as well as the rate of information gain explored throughout Chapter 4.

2.2.1 Motivations for kernel adaptation

The motivations for an addition of a dynamic landscape adaptation of the model are threefold. Firstly, the quantity of information gathered by the gaze trajectory must be accounted for, as this is the fundamental purpose of a visual search. Secondly, the regularity of information-seeking saccades during

visual search must be present, most importantly the recently discovered phenomena of strongly theta-rhythmic saccades explored further in Section 3.1, as well as the well-studied phenomenon of inhibition of return (IoR) [40, 150]. Thirdly, any functionally informative description of search dynamics must couple these search dynamics to the current information to model search through active vision [10, 53, 144, 151].

To couple gaze dynamics to information gained continuously, we examine the structure of visual information gathering. In this work the target distribution $\pi_t(\mathbf{x})$ will encode the belief in the presence a searched-for target object in 2D space, in a Bayesian sense. Thus, any information during exploration of non-target locations will cause a depletion of belief (probability mass) from the target distribution π_t locally. This is equivalent in the dynamical model to raising the potential landscape $V_t(\mathbf{x}) = -\ln \pi_t(\mathbf{x})$ locally by some kernel centred at the gaze position whose width represents the spatial acuity of information gathering. This gives a very well-motivated definition of continuous information gathering, as information is directly equated to the logarithm of probability in reducing uncertainty, and uncertainty in the location of a target is exactly $\pi_t(\mathbf{x})$ in our search task.

In this section we show that the following requirements are satisfied by the novel addition of a single mechanism to FNS dynamics to construct a full model of gaze dynamics:

1. *Updating search information.* The quantity of information gathered along the gaze trajectory must be tracked internally, as this is the fundamental purpose of a visual search.
2. *Saccadic regularities.* The model must reproduce the regularities in information-seeking saccades: most importantly the recently discovered strongly theta-rhythmicity of saccades (Section 3.1), and the well-studied phenomenon of inhibition of return (IoR) [40, 150].
3. *Coupling dynamics and searched information.* Any functionally informative description of search dynamics must couple those dynamics to the information gathered so far, completing the feedback loop of active vision [10, 53, 144, 151].

2.2.2 Developing the model kernel adaptation as information gathering

In this work the target distribution $\pi_t(\mathbf{x}) = e^{-V_t(\mathbf{x})}$ encodes, in a Bayesian sense, the belief that the searched-for target lies at position $\mathbf{x}$. Fixating a location $\mathbf{x}_0$ without finding the target is evidence against the target lying near $\mathbf{x}_0$, so considering some limited and spatially local information gathering during a fixation, belief must be depleted with according to some spatial kernel function $K(\cdot)$ as:

$$\pi_{t+\mathrm{d}t}(\mathbf{x}) \propto \pi_t(\mathbf{x})\, e^{-K(\mathbf{x}-\mathbf{x}_0)dt},$$

where $K(\mathbf{x}-\mathbf{x}_0) \geq 0$ is the log-evidence per unit time against discovering a target at $\mathbf{x}$ gathered during $\mathrm{d}t$, which is most informative at $\mathbf{x}_0$ where the most information may be gathered (see [35, 145]). The dynamics of $\mathbf{x}_t$ are then driven by the effective potential $V_t(\mathbf{x}) = -\ln \pi_t(\mathbf{x})$, so this multiplicative depletion of belief is an additive increment to the potential, $V_{t+dt}(\mathbf{x}) = V_t(\mathbf{x}) + K(\mathbf{x}-\mathbf{x}_0)dt$, up to a spatially uniform constant which normalises the distribution π_t and acts as a scalar gauge symmetry on the potential V_t.

Information gathering is therefore equivalent to raising the potential locally by the information gained at each location. This gives a well-motivated, continuous definition of information gathering, which turns out to be identical to an additive modulation of the potential surface which controls dynamics due to their shared link to an internal belief distribution π_t. The amount gathered at $\mathbf{x}$ depends on its displacement from the gaze position, because acuity falls off away from the fovea. Writing $K(\mathbf{x})\mathrm{d}t = k(\mathbf{x}-\mathbf{x}_0)\,\mathrm{d}t/\tau$ to modulate the timescale of information gathering using τ, and to use a dimensionless kernel function $k(\cdot)$, the instantaneous law of kernel dynamics for gaze at $\mathbf{x}_t$ is

$$\partial_t V_t(\mathbf{x}) = \frac{1}{\tau}\, k(\mathbf{x}-\mathbf{x}_t), \tag{2.5}$$

where $k(\cdot)$ is an approximately radially symmetric function that tends to zero for large arguments and is scaled so that $k(\mathbf{0}) = 1$. Throughout this work we use an isotropic Gaussian,

$$k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right). \tag{2.6}$$

the two parameters σ_k and τ control the spatial and temporal kernel dynamics respectively. The time constant τ is the time a gaze position needs to raise the potential by one nat at its centre, so $1/\tau$ is the peak rate of information acquisition. The width σ_k is the spatial scale of information gathering. We will see that τ is related to theta-rhythmic activity in gaze dynamics (Section 3.1). We can now show that σ_k is not a free parameter of our model but is set by the acuity of the human eye.

Relation of the kernel width to spatial acuity

The width σ_k is tied to how the detectability of the target falls with retinal eccentricity E, the angular distance from the point of gaze. This is quantified by the visibility map $d'(E)$, measured psychophysically for a given target and background, which ideal-searcher models [10, 35, 37, 144, 152] use to weight the evidence from each fixation in the Bayesian update of target belief [36, 37]. The visibility map declines with eccentricity because due to the decreased resolution outside the central fovea [3–5], through the fall-off of acuity, cortical magnification and crowding, with a half-resolution eccentricity E_2 of a few degrees (about 2.3° in Geisler and Perry [153]) [154, 155]. For an observer with sensitivity d', the expected log-evidence against a target at a target-absent location is $d'^2/2$ per fixation [36, 37], so the kernel is the normalised squared visibility map,

$$k(E) = \left[\frac{d'(E)}{d'(0)}\right]^2,$$

and σ_k is the width of this profile. Thus σ_k is determined by the acuity scaling of the eye and the spatial scale of the target rather than being a free parameter. Calculating the E_2 of our model as the characteristic scale of our Gaussian σ_k, we find a $E_2 = \sigma_k \div 36.22\,\mathrm{px}^{\circ -1} = (80 \div 36.22)^\circ \approx 2.20^\circ$, which is biologically accurate for such a search task as we consider.

2.2.3 Adaptive kernel dynamics as a driver of long-time rhythmic exploration

We now predict analytically the rate of excursions under these modelled dynamics in Equation (2.1) due to kernel deposition, whose mechanism is schematically shown in Figure 2.2**a)**–**d)**. The central idea to understand these kernel-driven transitions is to form a tractable transition condition: the effective potential well becomes concave-down at the gaze position. Such a well no longer localises the gaze position $\mathbf{x}_t$, and the Lévy-driven noise will then produce excursions to other modes.

We consider a single isotropic Gaussian mode of the target distribution, centred at zero with standard deviation σ_{fix}, and gaze position fixating within it. In terms of the radial coordinate r about the mode, the effective potential $V_t(r)$ therefore starts as the quadratic well

$$V_0(r) = -\ln \pi_0(\mathbf{x}) = \frac{r^2}{2\sigma_{\mathrm{fix}}^2}.$$

The choice of Gaussian loses very little generality, because a Taylor expansion will show that locally every mode is approximated to second order by a quadratic well—referred to as Laplace's approximation. We assume that the gaze position is localised near $r \approx 0$ and that $\sigma_k \gg \sigma_{\mathrm{fix}}$, so that the small noise-driven motion about the tight potential well with width σ_{fix} can be smoothed over, and kernel deposition can be assumed to be centred at $r = 0$. Excursions are then driven by the deterministic kernel dynamics alone. Large Lévy jumps will introduce Poisson-distributed transitions, as a result of the Lévy-Itô decomposition discussed in Section 1.2.3 [134], which will be arrhythmic with some mean value. We do not analyse this addition to rhythmic transitions between wells, but we discover this in the data in Chapter 3, and is shown for a number of kernel rates τ in Table 3.1. Under Equation (2.5)

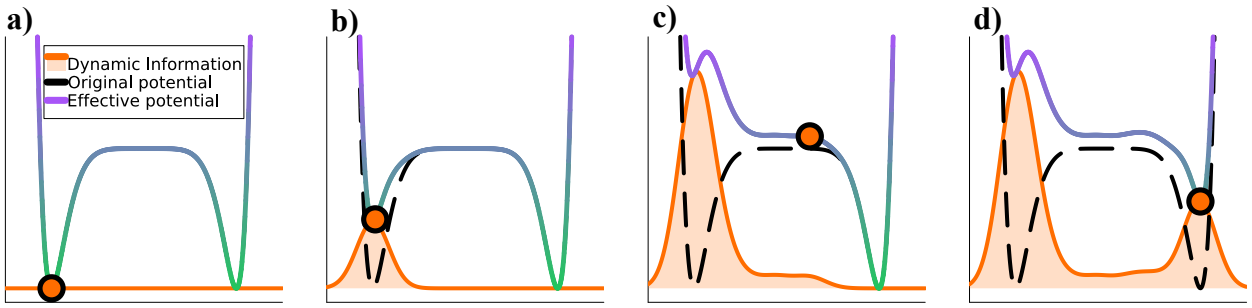


Figure 2.2: Schematic representation of kernel dynamics influencing an information-driven transition between potential wells. **a)** Two potential wells in a 1D potential landscape, with a high potential region separating them, which would represent low saliency in a model of gaze. $\mathbf{x}_0$ is positioned in the left well, which represents gaze position. **b)** Information acquisition as the kernel (orange) grows, gaining information and raising the potential landscape locally. **c)** A saccade-like transition, where the left potential well is inverted, followed by a transition to the right potential well. **d)** Exploration in the right well, with the information-gathering cycle resetting on a long t_θ-scale.

with the gaze at the mode centre,

$$\partial_t V_t(r) = \frac{1}{\tau} \exp\left(-\frac{r^2}{2\sigma_k^2}\right),$$

which integrates to

$$V_t(r) = \frac{r^2}{2\sigma_{\text{fix}}^2} + \frac{t}{\tau} \exp\left(-\frac{r^2}{2\sigma_k^2}\right). \tag{2.7}$$

The excursion criterion here is a negative radial second derivative of the explored potential at the fixation point $r = 0$, leaving the gaze unconstrained. The second spatial derivative is

$$\frac{\partial^2 V_t(r)}{\partial r^2} = \frac{1}{\sigma_{\text{fix}}^2} + \frac{t}{\tau\,\sigma_k^2}\left(\frac{r^2}{\sigma_k^2} - 1\right) \exp\left\{-\frac{r^2}{2\sigma_k^2}\right\},$$

and at $r = 0$ (where the Hessian is isotropic) it vanishes at the time

$$t_\theta = \tau\,\frac{\sigma_k^2}{\sigma_{\text{fix}}^2}. \tag{2.8}$$

For $t > t_\theta$ the origin is a local maximum of V_t, making this explored region very low interest and unlikely to be explored, which yields a dynamical origin of inhibition of return [40, 66, 150, 156–158]: the gaze is expelled from the location it has just exploited. Strictly, the criterion marks the onset of delocalisation; the excursion to another mode is then carried by the Lévy noise.

It is unsurprising that t_θ is proportional to τ. The dependence on the ratio of variances is of more interest. The dwell time is proportional to the variance σ_k^2 of the adaptive kernel and inversely proportional to the variance σ_{fix}^2 of the mode: a broad kernel takes longer to flatten a given well, and a sharply distributed fixation point will confine the gaze for longer. The result also extends beyond the Gaussian kernel, as σ_k represents the curvature of the kernel function at zero $k(\mathbf{0})$. An extension of the kernel method to functions which have a cusp at the origin may be biologically realistic [153–155], and could produce a rich description of the drifts and fixational movement of the eyes beyond tremor frequencies [41, 73], but such an approach is less tractable analytically and less stable computationally.

2.3 Local harmonic oscillations

One predictive feature of FNS dynamics is the local oscillations in sampling within confined potentials. We can quantify such high frequency oscillations under FNS dynamics by considering the deterministic part of our FNS equations of motion in Equation (2.1). Considering a Gaussian probability distribution of fixation positions $\pi(\mathbf{x}) \propto \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_{\text{fix}}^2}\right)$ and thus logarithmic gradients $\nabla \ln \pi(\mathbf{x}) = -\frac{\mathbf{x}}{\sigma_{\text{fix}}^2}$. Using this as our saliency landscape $\pi(\mathbf{x})$ without considering perturbation to the explored potential from the slowly updating kernel dynamics, our equations of motion are

$$\begin{aligned} d\mathbf{x}_t &= -\gamma c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt \end{aligned} \tag{2.9}$$

where we ignore the stochastic noise driving this equation to only consider the coupled ODE which describes deterministic part of $\mathbf{x} = \mathbf{x}_t$ and $\mathbf{p} = \mathbf{p}_t$, where the loss of subscript in typical when moving from an SDE to an ODE

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= -\gamma c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2} + \beta \mathbf{p}, \\ \frac{d\mathbf{p}}{dt} &= -\beta c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2} \end{aligned} \tag{2.10}$$

Assuming an underdamped regime where the system is periodic, corresponding to parameter range $\beta > \gamma\sqrt{c_\alpha}/\sigma_{\text{fix}}$, the resonant ordinary frequency $f = 2\pi\omega$ is given by

$$f = \frac{1}{2\pi\sigma_{\text{fix}}} \sqrt{\beta^2 c_\alpha - \frac{\gamma^2 c_\alpha^2}{4\sigma_{\text{fix}}^2}} \tag{2.11}$$

2.4 Lévy superdiffusion

To model the superdiffusion of a computational exploratory dynamical system such as found in searching eye movements, our system is driven by Lévy noise withing an over-damped Langevin regime (see Sections 1.2.1 to 1.2.3). Such underlying dynamics are known as Lévy flight (see 1.2.3) and are the paradigm superdiffusive system. However, both the potential-driven adaptive kernel dynamics, and auxiliary momentum dynamics add complexity to the superdiffusive properties of the model. Quantifying superdiffusion in this system exploring a confined and dynamic potential landscape requires a dedicated analysis of how Lévy dynamics will introduce monoscaling superdiffusion which is self-similar on all scales, with the same superdiffusive patterns driving large jumps across a landscape as those small-scale jumps when considering finer movements. Lévy motion unifies these scales under a single superdiffusive pattern, motivating our unification of gaze dynamics under a single model.

Model dynamics as a generalised Lévy-driven Langevin system

Lévy processes are the canonical form of monoscaling superdiffusion, yet the analysis of the precise form of diffusive scaling requires close analysis. Lévy flight superdiffusion is analysed from the deterministic dynamics of the probability distribution of jump size $P(\|\Delta\mathbf{x}\|)$ dynamics of the probability distribution. This problem of understanding the dynamics of the distribution of $\mathbf{x}$ is identical to that approached when studying classical Brownian motion where the classical Fokker-Planck equation was developed (see 1.2.1). Here in the case of a generalised Lévy-driven Langevin system with momentum coupling, the deterministic dynamics of the distribution of $\mathbf{x}$ is governed by the fractional Fokker Planck equation, as described in [33, 159, 160]. We can consider the phase space of a generalised coupled set of n stochastic differential equations of d-dimensions of $\mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2, \cdots, \mathbf{z}_n]^T$, where $d\mathbf{z}_t = (\mathbf{D}+\mathbf{Q})\, b(\mathbf{z}_t)dt + \mathbf{D}^{\frac{1}{\alpha}} d\mathbf{L}_t^\alpha$ for $(n \times d)$–dimensional Lévy noise. Here, $\mathbf{D}$ is the positive semi-definite matrix which defines strength and coupling of noise across $\mathbf{z}$, and $\mathbf{Q}$ represents the skew-symmetric

matrix defining dynamical coupling across $\mathbf{z}$. For such a system the fractional Fokker-Planck equation can be expressed exactly:

$$\partial_t P(\mathbf{z},t) = -(\mathbf{D}+\mathbf{Q})\nabla_{\mathbf{z}}\left[b(\mathbf{z})P(\mathbf{z},t)\right] - \mathbf{D}\mathcal{D}^{\alpha}_{\mathbf{z}}P(\mathbf{z},t) \tag{2.12}$$

where $\mathcal{D}^{\alpha}_{\mathbf{z}}$ represents the α-fractional Riesz derivative, which generalises the operation of the Laplacian to fractional orders, and is defined in Fourier space as $\mathcal{D}^{\alpha}_{\mathbf{z}}[f] = \mathcal{F}^{-1}\|\mathbf{k}\|^{\alpha}\mathcal{F}[f]$, with the case of $\alpha = 2$ restoring the ordinary Fokker-Planck equation with $\mathcal{D}^{\alpha=2}_{\mathbf{z}} = -\nabla^2_{\mathbf{z}}$. It can be shown that our model of gaze dynamics is recovered for $\mathbf{z} = [\mathbf{x},\mathbf{p}]$, $\mathbf{D} = \begin{bmatrix}\gamma & 0\\ 0 & 0\end{bmatrix}$ and $\mathbf{Q} = \begin{bmatrix}0 & -\beta\\ \beta & 0\end{bmatrix}$, shown in [33]. The $b(\mathbf{z})$ drift term required to recover a given distribution is presented in Equations (2.3) to (2.4). Any solutions to the dynamics in Equation (2.12) are nontrivial, as the fractional derivative results in a nonlinear multivariate integro-differential equation which is difficult to propagate, even numerically.

Analysis of Lévy flight superdiffusion from the fractional Fokker-Planck equation

When considering diffusion over short timescales, the physics of fractional Langevin systems with $\alpha > 1$ are noise dominated. In our Lévy driven system, short times are dominated by α-stable dL^{α}_t Lévy noise, across power-law distributed spatial scales of short jumps and nonlocal jumps to distant positions in short times. This is corroborated by the scaling of Lévy noise as $dL^{\alpha}_{\Delta t} = \Delta t^{1/\alpha} dL^{\alpha}_1$ in Equation (1.7) is stronger than ballistic effects which scale in short times as Δt. Thus, considering the fractional Fokker-Planck equation evolving from a delta function initial condition $P(\mathbf{z},0) = \delta^n(\mathbf{x}-\mathbf{x}_0)\delta^n(\mathbf{p}-\mathbf{p}_0)$ evolving through a quadratic potential using the position coordinate $\Delta\mathbf{x} = \mathbf{x}-\mathbf{x}_0$, yielding a drift term $\mathbf{b}(\Delta\mathbf{x}) = \mathbf{b}(\mathbf{x}_0) + \mathcal{O}(\Delta\mathbf{x})$. We can construct a leading-order short time solution up to $t = \Delta t$ and marginalise over the auxiliary momentum $\mathbf{p}$ which in practise is a nuisance variable, we have:

$$\partial_t P(\Delta\mathbf{x},t) = -\gamma\mathcal{D}^{\alpha}_{\mathbf{x}}P(\Delta\mathbf{x},\Delta t) - \nabla_{\mathbf{x}}\cdot\left[\left(\gamma\mathbf{b}(\Delta\mathbf{x}) + \beta\,\mathbb{E}[\mathbf{p}\,|\,\Delta\mathbf{x},\Delta t]\right)P(\Delta\mathbf{x},\Delta t)\right] \tag{2.13}$$

where we can truncate these terms to highest order using

$$\mathbf{b}(\Delta\mathbf{x}) = \mathbf{b}_0 + \mathcal{O}(\Delta\mathbf{x}), \qquad \mathbb{E}[\mathbf{p}|\Delta\mathbf{x},\Delta t] = \mathbf{p}_0 + \mathcal{O}(\Delta t), \qquad \mathbf{v}_0 = \gamma\,\mathbf{b}_0 + \beta\,\mathbf{p}_0 \tag{2.14}$$

which gives

$$\begin{aligned}
\partial_t P(\Delta\mathbf{x},\Delta t) &= -\gamma\mathcal{D}^{\alpha}_{\mathbf{x}}P - \nabla_{\mathbf{x}}\cdot\left[\left(\mathbf{v}_0 + \mathcal{O}(\Delta\mathbf{x}) + \mathcal{O}(\Delta t)\right)P\right]\\
\partial_t P(\Delta\mathbf{x},\Delta t) &= -\gamma\mathcal{D}^{\alpha}_{\mathbf{x}}P - \mathbf{v}_0\cdot\nabla_{\mathbf{x}}P + \left[\mathcal{O}(\Delta t^0) + \mathcal{O}(\Delta t^{1-1/\alpha})\right]\cdot P\\
\partial_t \hat{P}(\mathbf{k},\Delta t) &= -\gamma\|\mathbf{k}\|^{\alpha}\hat{P} - i\,\mathbf{v}_0\cdot\mathbf{k}\,\hat{P} + \mathcal{O}(\Delta t^0)\hat{P} \qquad (2.15)\\
\implies \hat{P}(\mathbf{k},\Delta t) &= \exp\left[-\gamma\|\mathbf{k}\|^{\alpha}\Delta t - i\mathbf{v}_0\cdot\mathbf{k}\,\Delta t\right]\left[1+\mathcal{O}(\Delta t)\right] \qquad (2.16)
\end{aligned}$$

Showing that, to first order in Δt, the short-time diffusion under our model Equation (2.1) yields a $\mathcal{S}\alpha\mathcal{S}$ distribution, defined through it's characteristic function, which is transported with velocity $\mathbf{v}_0 = \gamma\mathbf{b}_0 + \beta\mathbf{p}_0$.

Quantification of Lévy superdiffusion monoscaling

Following on from the solution for the characteristic equation $\hat{P}(\mathbf{k},\Delta t) = \exp\left[-\gamma\|\mathbf{k}\|^{\alpha}\Delta t - i\mathbf{v}_0\cdot\mathbf{k}\,\Delta t\right][1+\mathcal{O}(\Delta t)]$, we reduce to the characteristic function of Fourier-space jump sized $k = \|\mathbf{k}\|$. For $0 < q < 2$ this change of variables will give the relation [161]

$$\langle\|\Delta\mathbf{x}\|^q\rangle = C_q\int_0^{\infty}\left[1 - \mathrm{Re}\,\hat{P}(k,\Delta t)\right]k^{-1-q}\,\mathrm{d}k, \qquad C_q = \frac{q(1-q)}{\Gamma(2-q)\cos(q\pi/2)}, \tag{2.17}$$

where C_q is defined at $q = 1$ by its limit, $C_1 = 2/\pi$. Substituting our result of Equation (2.16) into Equation (2.17) and evaluating after a further change of variable $u = \gamma\,\Delta t\,k^{\alpha}$, we have for

$0 < q < \alpha$:

$$\langle \|\Delta \mathbf{x}\|^q \rangle = C_q \left[\frac{\Gamma\left(1 - \frac{q}{\alpha}\right)}{q} (\gamma \Delta t)^{q/\alpha} + \frac{\Gamma\left(\frac{2-q}{\alpha}\right)}{2\alpha} \frac{v_0^2}{\gamma^2} (\gamma \Delta t)^{(2-\frac{2}{\alpha})+\frac{q}{\alpha}} + \mathcal{O}\left(\Delta t^{1+q/\alpha}\right) \right]. \tag{2.18}$$

where the first term is the moment of an α-stable Lévy flight with scale $(\gamma \Delta t)^{1/\alpha}$, which is finite only for $q < \alpha$ [162]. The second term is the ballistic contribution of the momentum, which will tend to dominate as oscillations over longer times. Since $2 - \frac{2-q}{\alpha} > \frac{q}{\alpha}$ for $\alpha > 1$, it is subleading at short times, effecting dynamics more as $q \to 0$ and $\alpha \to 1$. This gives diffusion through fractional absolute moments scaling as

$$\langle \|\Delta \mathbf{x}\|^q \rangle \propto \Delta t^{\lambda(q)}, \qquad \lambda(q) = \frac{q}{\alpha}, \qquad 0 < q < \alpha. \tag{2.19}$$

Thus, for $q < \alpha$ the modelled dynamics of Equation (2.1) reproduce the scaling of a Lévy flight [143].

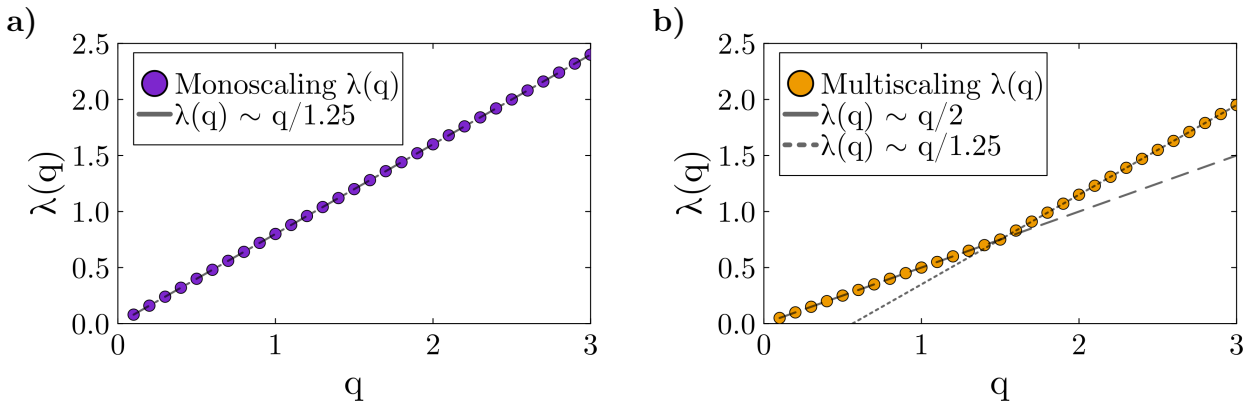


Figure 2.3: Moment spectra $\lambda(q)$ distinguish mono- and multiscaling superdiffusion. The spectrum is defined by $\langle \|\Delta \mathbf{x}\|^q \rangle \sim \Delta t^{\lambda(q)}$, with q the moment order (Equation (1.6)). **a)** Monoscaling: $\lambda(q)$ is linear in q. For a Lévy flight with stability index α, $\lambda(q) = q/\alpha$ for $q < \alpha$ (Equation (2.18)). Both gaze data and our model are monoscaling up to $q = 3.0$, implying a unified process driving motion across spatial scales (Section 3.3). **b)** Multiscaling (strong anomalous diffusion, SAND): $\lambda(q)$ is piecewise linear, with a slope change separating small displacements (low q, here normal diffusion) from large ones (high q, here superdiffusion). This is typical of a Lévy walk or of local Brownian motion punctuated by rare large jumps [124], the standard model of gaze as fixation punctuated by rare saccades.

Such superdiffusion will produce a self-similar process with a single scaling exponent $1/\alpha > 1/2$ across all scales. This is called monoscaling superdiffusion, where superdiffusion is present and identical across all scales of q.

However, for $q \geq \alpha$ the moments of an unbounded Lévy flight diverge and the expansion above no longer applies. It is thus important to computationally examine the behaviour of a Lévy flight during exploration of particular potentials, whether or not they are unbounded or dynamic, and if any post-processing is done to handle divergences in the variance of jumps. In other studies, Lévy flights have been found to have $\lambda(q) \neq \frac{q}{\alpha}$, and are thus proposed to be multifractal with different scaling behaviour for large jumps as for small. This is the signature of strong anomalous diffusion [163], however it is misleading, as a Lévy flight is defined by monoscaling self-similarity. In this study, we find that under the confined dynamics and light low pass filtering to produce continuous trajectories, the Lévy flight yields a single monoscaling superdiffusive exponent of $\lambda(q) = \frac{q}{\alpha}$

In summary, this chapter presents three key dynamical components of the unifying model of gaze dynamics we present: a kernel which couples regular information gathering with dynamical transitions, the high frequency oscillations which drive fine spatial dynamics, and superdiffusive spatial dynamics which are defined by a single exponent across all spatial scales. In Section 2.1 we compared the

modelled setup with an empirical experiment performed in [61]. To examine this model as a valuable predictive and explanatory model of gaze dynamics, we compare to these human data.

Chapter 3: Empirical Validations

In this chapter, we evaluate the predictions of the model developed in Chapter 2 against high precision eye-tracking data from Näher et al. [61], described in Section 2.1. Motivated by established spatiotemporal signatures of gaze during visual search, we examine three key dynamical features: (1) theta-rhythmic saccades, (2) gamma-scale 40–100 Hz fixational ocular drift, and (3) global superdiffusion across spatial scales [56, 164]. Our unifying framework seeks to explain these phenomena through network-level neural activity, including the proposed connection between microtremor and neural gamma, theta-gamma coupling, and the monoscaling of superdiffusion across spatial scales.

Section 3.1 examines theta rhythmicity of saccades, comparing the model's transitional rhythmicity with the saccade timing in the data. Section 3.2 examines gamma-scale oscillations during visual fixations, testing the model's local high-frequency harmonic oscillations against fixational drift. Section 3.3 examines superdiffusion in search, assessing whether the global spatial statistics of gaze are consistent with Lévy superdiffusion. Finally, Section 3.4 presents the spatiotemporal predictions of the model across these scales and discusses how they unify the experimental and modelling findings above.

3.1 Theta rhythmicity of saccades

The approximate 200 ms timescale of human saccades has long been understood, but their rhythmicity is not often quantified beyond this approximate scale [25, 165–167]. Direct measures of saccade rhythmicity are found in the statistics of the distribution of inter-saccadic intervals (ISIs), the interval durations between saccades Δt_i. One such dimensionless measure is the variance of the ISI distribution in log-space. Using this, we can construct a measure called the rhythmicity index (RI):

$$\mathrm{RI} = \frac{1}{\mathrm{CV}_{\ln}}, \qquad \mathrm{CV}_{\ln} = \sqrt{e^{\sigma_{\ln}^2} - 1} \tag{3.1}$$

where $\sigma_{\ln}$ is the standard deviation of the distribution of the natural log of ISIs $\ln \Delta t$, with $\mathrm{CV}_{\ln}$ often called the geometric CV. This RI is an effective measure of the spread of a strictly positive distribution. A Poisson distributed variable will give $\mathrm{RI} = \frac{\sqrt{6}}{\pi} \approx 0.7800$[1], and it is found that for an ISI distribution gathered from saccades across a disordered grid of dots is $\mathrm{RI} = 2.69$ [61].

Strong rhythmicity of saccades with a slowly changing theta rhythm

The recent discovery of the strength of saccade rhythmicity by Näher et al. [61] required an accounting of the variable rhythms of biological systems. The timescales of the brain are not driven by a perfect clock, but instead from characteristic durations of certain regular patterns. As such, the 200 ms estimate for saccades during a visual search is not expected to be a universal timescale across subjects and trials. In the case of saccades, this rate can variabley be found in the theta-band of frequencies 4–8 Hz (see Section 1.1.2). This slowly varying theta rate is expected to be stable within a single individual and a single trial, and thus we consider a local measure of rhythmicity between the intervals between saccades, called inter-saccadic-intervals (ISI). Such a metric can be proposed through the local rate-corrected rhythmicity index (RCRI) [61]:

$$\mathrm{RCRI} = \frac{1}{1 + \langle \mathrm{CV}_2 \rangle}, \qquad \mathrm{CV}_2(i) = 2\frac{\|\Delta t_{i+1} - \Delta t_i\|}{\Delta t_{i+1} + \Delta t_i} \tag{3.2}$$

which allows quantification of purely pairwise rhythmicity, comparing each interval Δt_{i+1} to it's predecessor Δt_i, and averaging across all intervals i considered. To give baseline values, the RCRI of a Poisson process is $\mathrm{RCRI} = 0.5$, and a perfectly rhythmic process will give $\mathrm{RCRI} = 1$. The use of RCRI

[1] This can be found from the exponentially distributed intervals of a Poisson process with mean λ as $X \sim \lambda e^{-\lambda X}$, changing variables to find the distribution of $\ln X = Y \sim \lambda \exp[Y - \lambda \exp Y]$. This distribution will have $\sigma_{\ln} = 0.986$

as a local metric for rhythmicity has produced two results. Firstly, human saccades are very strongly rhythmic, with RCRI ≈ 0.765 for saccades exploring a disordered grid of dots [61]. Secondly, it has been found that the instantaneous saccade rate is a strong predictor for the local saccade rhythmicity RCRI, with a Pearson correlation found to be $\rho = 0.39, P = 0.001$, with saccade rate acting as a significant predictor of RCRI with $\beta = 0.0114\,\mathrm{Hz}^{-1}, P = 0.014$ [61]. It has been found that those trials binned into the low-, medium-, and high- saccade rate tertiles will show a positive correlation of saccade rate with higher RCRI [61], as shown in Figure 3.1**b)**. A table comparing these indices and rhythmic patterns is also shown below in Table 3.1

Model				**Data**			
Saccade Rate	**RCRI**	**RI**	**⟨ISI⟩**	**Saccade Rate**	**RCRI**	**RI**	**⟨ISI⟩**
$\mathbf{t}_\theta = 213.33\,\mathrm{ms}$	0.6787 [0.675–0.683]	2.116 [2.08–2.15]	195.0 ms	Low	0.73	2.76	210 ms
$\mathbf{t}_\theta = 177.78\,\mathrm{ms}$	0.6857 [0.685–0.692]	2.237 [2.20–2.28]	178.4 ms	Medium	0.76	3.43	235 ms
$\mathbf{t}_\theta = 142.22\,\mathrm{ms}$	0.6990 [0.696–0.702]	2.364 [2.33–2.40]	152.7 ms	High	0.78	3.60	260 ms

Table 3.1: Effect of saccade rate on rhythmicity and ISI peak. The left four columns represent the rhythmicity found in the modelled saccades, with saccade rate modulated with different values of kernel timescale τ. Along with standard parameters, as described in Section 2.1, the kernel timescales are $\tau = 2.0\,\mathrm{ms}, 2.5\,\mathrm{ms}, 3.0\,\mathrm{ms}$. Histograms of the ISIs for these simulations are shown in Figure 3.1**a)**. We reproduce the result that high saccade rates are correlated with high rhythmicity [61], as seen in monotonically increasing RCRI and RI with increasing saccade rate. Empirical results are given in the right four columns [61].

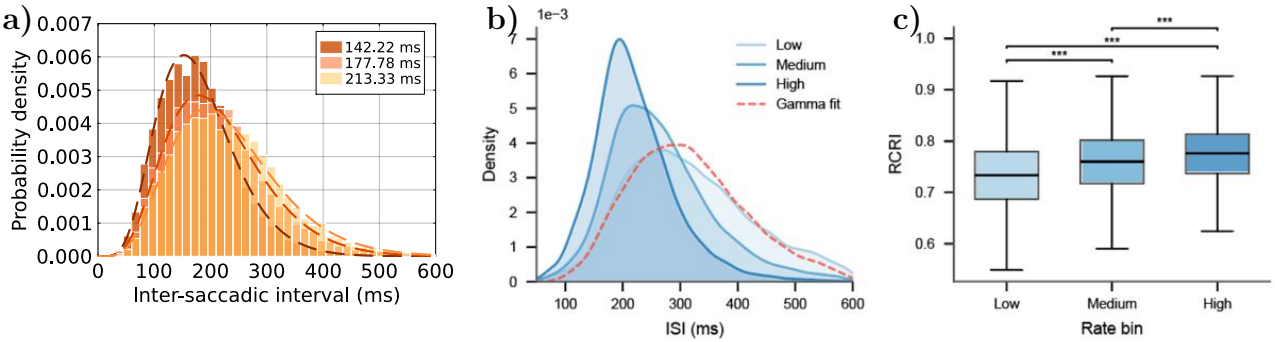


Figure 3.1: **(a)** The distribution of ISIs from the model given different underlying theta rhythms $t_\theta = 142.22\,\mathrm{ms}, 177.78\,\mathrm{ms}, 213.33\,\mathrm{ms}$. Full statistics and rhythmicity indices are given in Table 4.1. Gamma fits have shape parameters $k = 6.53, 5.86, 5.13$ respectively, demonstrating that saccade rate and rhythmicity are correlated. **(b)** Empirical distribution of ISIs after trials are binned into low-medium- and high-tertiles. **(c)** Trials binned with shorter ISIs are correlated with a high rhythmicity of saccades ($\rho = 0.39, P = 0.001$). Such a conclusion corresponds to the kernel dynamics presented, where faster kernel dynamics will increase rhythmic transitions (see Section 2.2.3). ***(b)*** *and* ***(c)*** *adapted from Figure. 5 of Näher et al. [61]*

Interpretations of local rhythmicity and saccade rate for a model of gaze dynamics

These results support our unified framework of gaze dynamics, specifically those predictions of the kernel dynamics described in Section 2.2.3. We have shown good agreement between the tight gamma renewal fit of modelled ISIs and the empirical ISI statistics. We have also introduced a natural parameter τ with which to modulate the slowly varying theta-scale saccade rate, whose effects on rhythmicity and average saccade rate are shown in Table 3.1. A correlation between high saccade rate and high rhythmicity is predicted from the origin of regular dynamical-informational transitions as described in Section 2.2, where a stronger kernel produces more rhythmic transitions, and a weaker kernel will approach an arrhythmic Poisson point distribution of ISIs, as fluctuation-driven exploration begins to dominate the search, reducing rhythmicity in saccades.

3.2 Gamma oscillations during visual fixation

We next verify the short-time and fine spatial scales of our model predictions, specifically of those high frequencies found in local sampling of potential modes in Section 2.3. Typically short-time gaze dynamics are divided into microsaccades, low-frequency linear drift, and high frequency ocular tremor [41]. There is increasing evidence that microsaccades—which is the name given to saccades which do not interrupt fixation—are driven by the same statistics as regular (called macro-) saccades. Unifying and treating these microsaccades as unified with global macrosaccades is performed in our analysis of superdiffusion in the next section (Section 3.3). The next component of fixation, linear drifts have been found to be important in gathering information. An account of low-frequency linear drift fits into the model presented here, through similar local self-avoidance arising from kernel dynamics described in Section 1.1.1 [7].

But our focus here is that of the high frequency motion called ocular microtremor (OMT) found in fixation [67, 168, 169] and often treated as muscular noise [67, 72, 170]. In this thesis, we make the well-motivated prediction from our unified framework that OMT is the gamma-scale sampling pattern driven by neural gamma in visual search. In this section, we describe how OMT has previously been characterised (Section 3.2.1), comparisons of our modelling framework of neural gamma to the data (Section 3.2.2), and briefly discuss their implications for the study of OMT.

3.2.1 Previous treatment of ocular microtremor (OMT)

OMT is the smallest-amplitude component of fixational eye motion. It has been observed across experimental studies, with reported amplitudes of up to 30–40 sec of arc [58, 171]. Despite its small amplitude, OMT has been proposed to contribute to the dynamic sampling of high-contrast features and fine spatial detail [27, 70–72]. However, a recent physical study concluded that its magnitude may be insufficient to produce a meaningful change in the retinal image during prolonged fixation [172]. We therefore consider our results under the hypothesis that the role of microtremor will be task-dependant: it may play an important functional role in controlling gaze during visual search, when gaze is rapidly adjusted in response to incoming information, than during the prolonged fixation tasks where OMT is usually studied [57, 58, 171]. Further, the frequency and amplitude of OMT has been demonstrated to correlate with broader brain function and state, such as being a biomarker for neurological fatigue [173], anaesthesia [174–176], and schizophrenia [177].

In our search-task data, the tremor spectrum has a broad peak at a lower frequency than is typically reported. Our results yield a peak at 56 Hz, and with a larger estimated amplitude. By comparison, studies of longer fixation periods generally report more localised peaks around 80–90 Hz [67]. This reiterates the result that OMT is task-dependant and provides an external summary of computation in the visual system. In gathering our results, the present analysis differs from the existing literature in two important respects:

1. We examine eye movements during a naturalistic visual search task. The major studies which report OMT, as surveyed in the review by Graham et al. [67], instead measure tremor across a prolonged fixation, typically lasting more than 3000 ms [57, 58, 171].

2. The data considered here are positional, not accelerometer or direct velocity data. OMT is studied using velocity or acceleration, as these data effectively ignore the low frequency drifts and are more sensitive to OMT amplitude. In our data, the estimated tremor amplitude is larger than values reported during prolonged fixation: approximately 200 sec of arc, compared with values of up to 40 sec of arc in prior studies [58, 171] (see Appendix C for the decomposition of a typical fixation into Fourier components and their residual).

3.2.2 Analysis of fixational gaze velocity

We analyse the profile of high frequency components of fixational gaze trajectories during visual search through their velocities. The shorter, naturalistic searching fixations in our data pose a challenge for

estimating high-frequency motion. In our data, fixations have an average duration of approximately 200 ms, which will yields only ten cycles of a 50 Hz signal. Moreover, estimating high-frequency motion from noisy position measurements can be misleading, as the low frequencies dominate the positional signal power of a fixation [57, 58]. This motivates our data pre-processing to low-pass filter the instrumental noise, and numerically differentiate to consider the velocity instead. Such a method has been used elsewhere to analyse OMT, with signal amplitude found within instrument error [57].

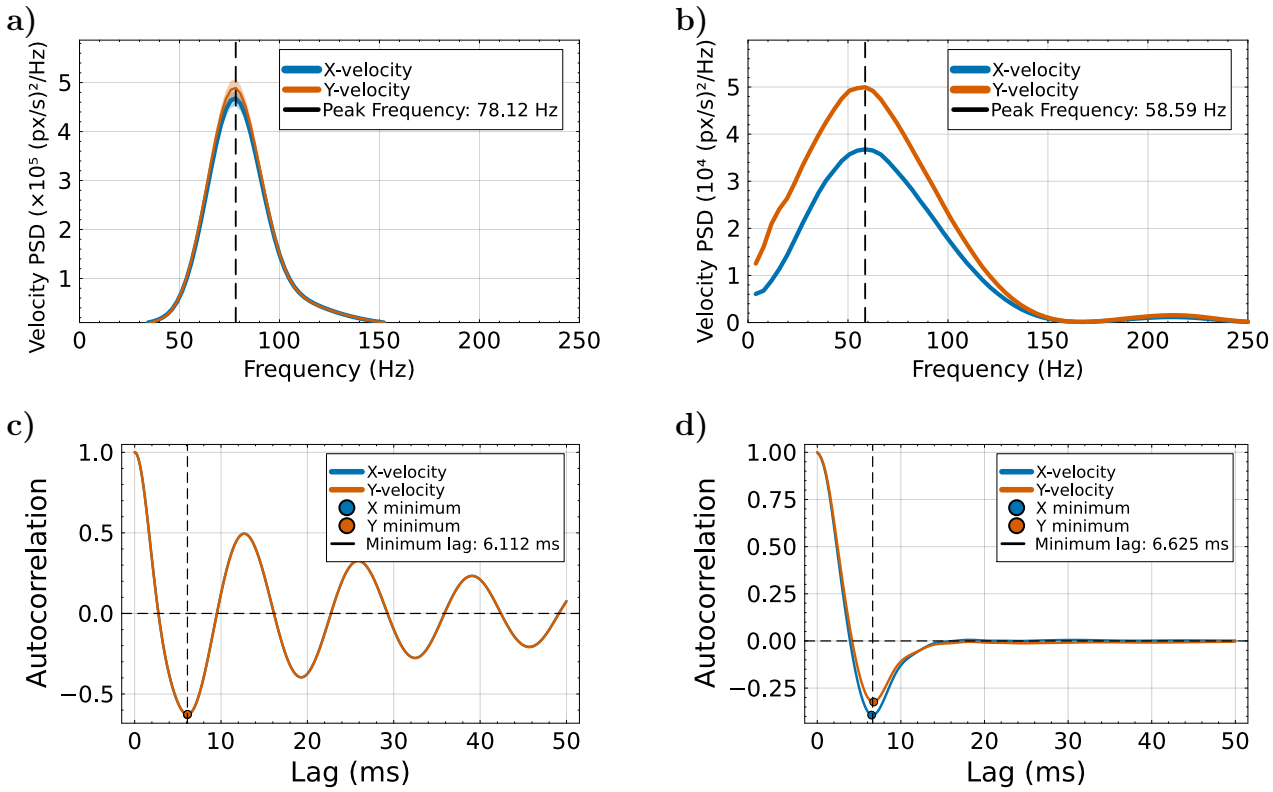


Figure 3.2: Velocity power spectra and autocorrelation of fixational eye movements in modelled and human gaze trajectories. **a)** Velocity power spectral density (PSD) of fixations in the modelled gaze trajectories, showing a broadband peak at $f = 78.12$ Hz with near-identical X and Y components. **b)** Velocity PSD of fixational periods in the human gaze data of [61], showing a broader peak at $f = 58.59$ Hz. In both a) and b), shaded ribbons denote confidence intervals and the dashed line marks the peak frequency. **c)** Velocity autocorrelation of the modelled trajectories, showing sustained oscillations. Taking the lag of the first minimum, $\tau_{\min} = 6.112$ ms, as a half period gives $f = 81.81$ Hz, in good agreement with the spectral peak in a). **d)** Velocity autocorrelation of the human data, which decays within approximately 20 ms. The first minimum, $\tau_{\min} = 6.625$ ms, gives $f = 75.47$ Hz. Pre-processing and spectral estimation are described in Section 3.2.2.

Data processing

The analysis pipeline was identical for the human eye-tracking data and the model gaze trajectories unless otherwise stated. Fixations were taken from the existing labels in the eye-tracking data or, where unlabelled, detected with a variant of the median-filter method of [178, 179], manually verified and implemented in `VisualFNS_Analysis.jl` (Appendix B). Fixations were truncated at the first NaN sample, and those shorter than 100 ms were discarded. To suppress high-frequency instrumental noise near the Nyquist frequency ($f_{\text{Nyquist}} = 250$ Hz), which numerical differentiation would otherwise amplify, positions in both datasets were low-pass filtered with a fourth-order Butterworth filter (200 Hz cutoff). Two-dimensional velocity was estimated by central differences of the filtered positions and, for the eye-tracking data, averaged across both eyes. Power spectral densities were computed with Welch's method using a Hann window of 256 ms (256 samples) and an overlap of 64 ms (64 samples). The resulting spectra are shown in Figure 3.2, with ensemble sizes and confidence ribbons.

Importance and discovery of gamma-scale fixational dynamics

Velocity power spectra of fixational periods show a broadband peak in both the modelled and human data (Figure 3.2**a,b)**). The modelled trajectories peak at 78.12 Hz with a narrow peak and near-identical X- and Y-velocity spectra (Figure 3.2a). The human data peak at 58.59 Hz with a markedly broader peak. The peak power of the human data is roughly an order of magnitude below that of the model. We can also extract velocity autocorrelation functions for the data and model, shown in Figure 3.2**c,d)**. This gives an independent data point which estimates the typical fixational reversal time, which we model through a harmonic oscillation. The first minimum in the autocorrelation, $\tau_{\min}$, is then interpreted as a half period ($f = 1/2\tau_{\min}$). The modelled velocity autocorrelation shows damped oscillations over several cycles, with $\tau_{\min} = 6.112$ ms and then $f = 81.81$ Hz, in good agreement with its spectral peak, as shown in Figure 3.2**c)**. The human velocity autocorrelation instead shows a single negative lobe (minimum ≈ -0.35 to -0.4) that decays to zero rapidly, within approximately 20 ms. The human data thus give $\tau_{\min} = 6.625$ ms and hence $f = 75.47$ Hz (Figure 3.2**d)**. The 17 Hz offset between this estimate and the human spectral peak is consistent with a broadband, weakly coherent process, in contrast to the narrowband, coherent oscillation of the model.

The human spectral peak lies well below the 80–90 Hz range that generally characterises mean tremor frequency in the literature, with a slightly larger amplitude. This may indicate different neural dynamics during visual exploration than during prolonged fixation. This interpretation finds a parallel in the finding that cortical gamma frequency is increased when viewing local regions of high contrast [26, 180, 181], where prolonged fixation may correspond to a tightly constrained control of the eye, analogous to exploring a visual location with high contrast. The broad power spectrum shown in Figure 3.2**b)** and the quickly attenuated velocity autocorrelation in Figure 3.2**d)** are interpreted as the influence of changing gamma rhythms of OMT across trials and subjects, as well as the effects of other high amplitude components of fixation, drift and microsaccades, which will broaden the spectral decomposition of fixations. Addressing these factors in designing further experiments is considered in Section 3.4

3.3 Search superdiffusion

Superdiffusion is often a prominent pattern in natural dynamical systems, where its existence implies some self-similar fractal structure. Such dynamics have not been the focus of other studies, and often visual trajectories have been considered as a fixational Brownian motion externally interrupted with saccades. We here present novel multiscale superdiffusive pattern in the diffusion of eye movements from eye trajectory data. Many studies which have described eye movement dynamics as normally diffusive stems from the sensitivity of data processing choices in finding anomalous diffusion exploring confined boundaries such as fixations. Conventional analyses often inadvertently obscure critical spatiotemporal features.

The most important methodological choice in accurately characterising visual diffusion patterns is the choice to time-average displacements, and choosing a short timescale over which diffusion scaling is considered. As demonstrated by Vilk et al. [120], aggregating data across disparate temporal regimes can wash out localised scaling behaviours, yielding an artefact of normal diffusion. Studies which construct regimes of normal and subdiffusive regimes are thus common when examining eye movements, but this ignores the continuous-time superdiffusion which we unify under multiple scales. For such studies, see [22, 44, 72, 99].

Analysis of multiscale superdiffusion in eye-tracking data

We analyse the binocular-averaged gaze trajectories over all 22 human subjects provided from the study in Näher et al. [61], as described in Section 2.1. From the onset of the search task, we time-average the fractional absolute displacements $\langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle$ for 30 values of q linearly separated from $q = 0.1$ to $q = 3.0$. These fractional absolute displacements are evaluated across all valid time lags $\Delta t < 1000$ ms with a resolution of $\Delta t = 1$ ms. All intervals containing NaN values (such as blinks) are

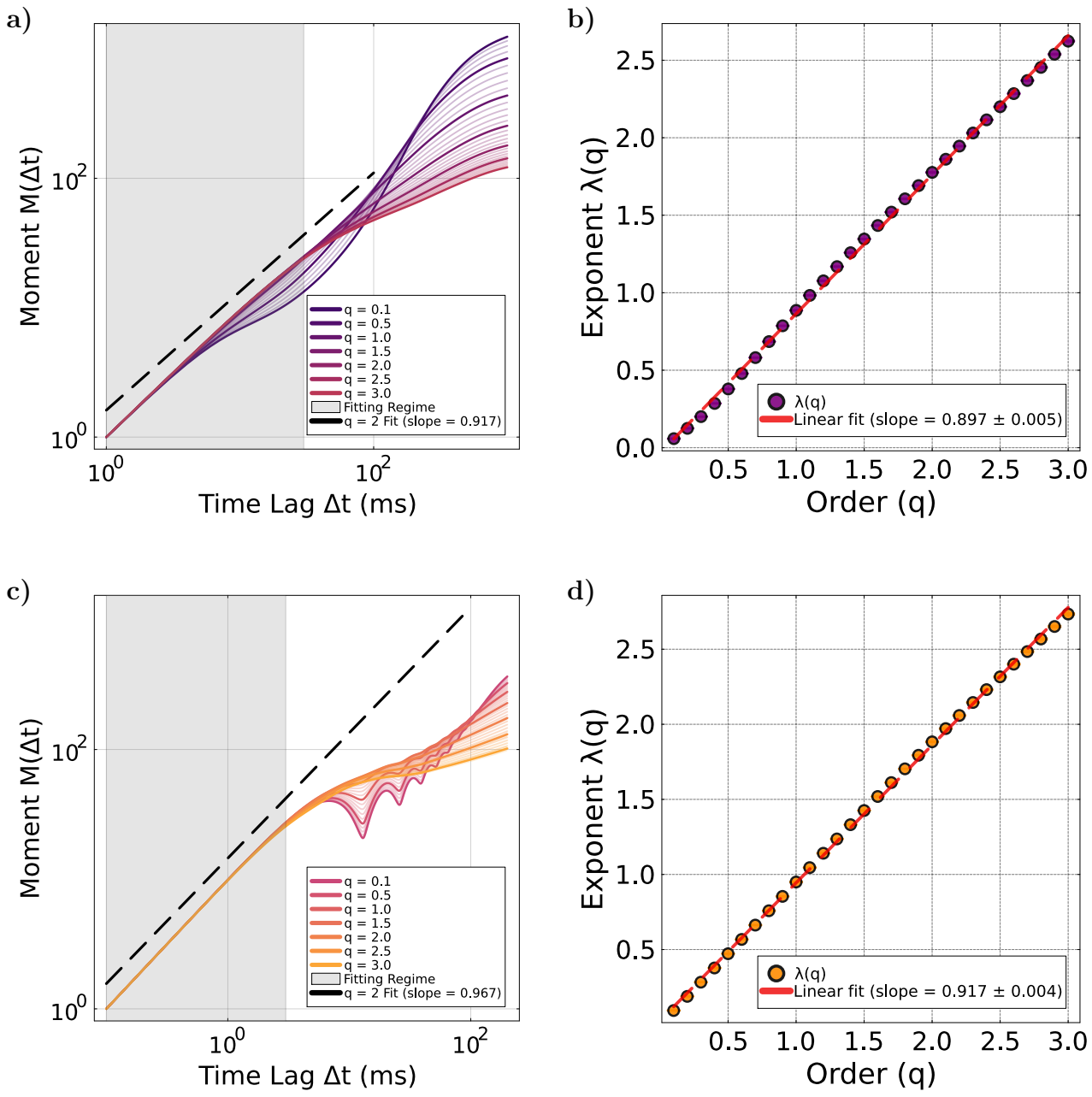


Figure 3.3: The model reproduces the monoscaling superdiffusion of human visual search through a linear spectrum of moments. Rescaled moments $M_q(\Delta t)$ (Equation (3.3)) and fitted scaling exponents $\lambda(q)$ for 30 values of q spaced linearly from 0.1 to 3.0. **Top row:** binocular-averaged gaze data from the 22 subjects of Näher et al. [61] (1846 trajectories, mean duration 5 s). **Bottom row:** 500 trajectories simulated from the model of Chapter 2 (mean duration 15 s). **(a,c)** Moments on log–log axes. The shaded region is the fitting regime and the dashed line shows the $q = 2$ fit, which recovers the square root of traditional MSD scaling. **(b,d)** The spectra are linear, $\lambda(q) = q/\alpha$, in both cases: $\alpha = 1.120 \pm 0.007$ for the data ($R^2 = 0.999$) and $\alpha = 1.091 \pm 0.005$ for the model, showing robust superdiffusion across scales, compared with $\alpha = 2$ recovering Brownian motion.

discarded. We then ensemble-average these displacements across all subjects.

We perform a linear best-fit in log-log space of the fractional moments $M_q(\Delta t)$ adefined in Equation (3.3) to find short-time power-law diffusive scaling with time, thus sampling from our moment spectrum function $\lambda(q)$ as defined in Equation (1.6). Each of these fractional moments are shown in Figure 3.3**a,c)**. The moment spectrum function $\lambda(q)$ thus is found to have a good fit to $\lambda(q) = (0.893 \pm 0.006)q$ with an error in the fit of $R^2 = 0.999$. This behaviour gives monoscaling across the shortest to longest spatial scales of fixations to saccades. The corresponding Lévy exponent given such linear monoscaling in $\lambda(q)$ is thus found to be $\alpha = \frac{q}{\lambda(q)} = 1.120 \pm 0.007$.

3.3.1 Spectrum of fractional moments

Figure 3.3 summarises the scaling of the time-averaged fractional absolute displacements. For comparison of the various scaling exponents, we define a rescaled moment function $M_q(\Delta t)$ to give good comparison of the spectrum of moments. $M_q(\Delta t)$ is here defined as

$$M_q(\Delta t) = M_q(\Delta t_{\min})^{-1} \langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle^{1/q} \sim \Delta t^{\lambda(q)/q} \tag{3.3}$$

with C_q defined such that $M_q(\Delta t_{\min}) = 1$ to give a shared origin at the first time lag Δt. Small q gives strong weight to the small, frequent displacements of fixational motion, while large q weights the rare, large displacements which are given in saccades. The shape of $\lambda(q)$ therefore tests whether one exponent may describe superdiffusion on all scales. A linear spectrum, $\lambda(q) = q/\alpha$, is monoscaling: all moments grow with the same exponent, as for a self-similar Lévy process with stability index α. A nonlinear spectrum is the general case found in many systems called strong anomalous diffusion which consist of two dynamical strategies stitched together at different scales [120, 124, 163, 182, 183]

Section 3.3 shows the rescaled moments $M_q(\Delta t)$ for the human eye-tracking data. Under this rescaling, the slope of each curve in log–log space is $\lambda(q)/q$. Monoscaling therefore predicts that these curves share a single slope across the whole spectrum of q, with $\frac{\lambda(q)}{q} = 1/\alpha$. For comparison, Brownian motion ($\alpha = 2$) has slope $1/2$ and ballistic motion has slope 1 across the entire spectrum q. The monoscaling exponent is found by performing a linear regression to our spectrum $\lambda(q)$. Section 3.3 shows these fitted exponents $\lambda(q)$ against q. The linear fit is good, with $\lambda(q) = (0.893 \pm 0.006)\, q$ and $R^2 = 0.999$, across the whole considered range $0.1 \leq q \leq 3.0$. This gives a Lévy index $\alpha = q/\lambda(q) = 1.120 \pm 0.008$, which is well below the Brownian value of 2 and so indicates strong superdiffusion. Because the spectrum is well described by a linear regression, there is no crossover in scaling between the low-q moments that probe fixational motion and the high-q moments that probe saccades. A single exponent describes both, validating our unifying model across both scales of study in gaze dynamics, derived naturally from FNS dynamics.

Implications of monoscaling superdiffusion

The most important implication of this particular superdiffusion results is that the strong monoscaling of visual search trajectories with Lévy index $\alpha = 1.120$ is robustly multiscale. The robust linear scaling of $\lambda(q) = \frac{q}{\alpha} = \frac{q}{1.120}$ across a spectrum of q-values $0 < q < 3$ is important, as it suggests/indicates/points to an identical process driving both fixational eye movements — including tremor, drifts, and microsaccades — with small relative values of $\|\Delta \mathbf{x}\|$ emphasised with low values of q, as well as large nonlocal saccades with large relative values of $\|\Delta \mathbf{x}\|$, emphasised through large values of q. Importantly, this validates our construction of a continuous-time unified model of eye movements, unified across all spatiotemporal scales of visual search.

3.4 Spatiotemporal predictions from the model

Throughout this chapter, we present a novel physics-inspired analysis of a standard gaze-tracking experiment during visual search which yields new results and predictions of gaze dynamics. In particular, the approximate timescale of saccades has been known since the beginning of the field of eye movement research, whereas the significance of the rate-corrected local rhythmicity is a very recent

result only discovered in the year of this present work. OMT as a phenomenon has been found in the literature [67], however no consensus exists as to it's importance in the functioning of gaze, and it has never been linked to a computational role of gamma-scale rhythms. Lastly, the mathematical analysis of search superdiffusion of gaze has only been recently undertaken in the literature, and without the well-motivated link to a model of gaze we present here. However, another benefit of comparing the generative unifying model presented in Chapter 2 to the data is that further dynamical predictions outside of the present literature can be made. Thus, we present experimental paradigms and features of this model which are being developed to better understand visual searching.

3.4.1 Saliency-frequency coupling in contrast-dependant search tasks

OMT frequency is here hypothesised to be the result of a harmonic oscillations in the control of the path of the eye during fixation tasks. An experiment may be designed in which targets with various degrees of contrast are presented to be searched. High contrast, localised points, such as tight, high spatial frequency Gabor patches, would produce localised saliency wells with low σ_{fix}. Whereas, low contrast, large low spatial frequency patches would produce saliency wells with a large width σ_{fix}. High contrast wells would correspond to shorter theta timescales, quadratic in the effective saliency well size as in Equation (2.8), and a microtremor rhythm which is approximately inverse to the effective saliency well size as in Equation (2.11). The effect of visual contrast has been found to be significant in gamma frequency in the visual system, and this quantitative relationship provides an experimental prediction of the relationship between three data points: $t_\theta, f_{\text{peak}}$, and k_{Gabor}. A simple regression analysis can be achieved here to find correlations between spatial frequency k_{Gabor} and the visual theta-rhythm t_θ and the f_{peak}. It can then also be tested if these effects are consistent with a potential well of variable size, such that $t_\theta \propto \sigma_{\text{fix}}^{-2}$ and $f \propto \sigma_{\text{fix}}^{-1}$

3.4.2 Smooth pursuit predictions and parametric-fitting techniques

Another experimental paradigm which provides insight into the behaviour of the control of visual trajectories. In particular, the exact same model can be examined in the behavioural paradigm of smooth tracking of a moving object. In particular, a typical experiment would be the tracking of a smoothly moving pendulum. It is realistically found that a typical eye movement trace contains discontinuous micro-saccades during smooth pursuit, and discontinuous catch-up saccades. In particular, we expect from the model that the instability will be most present in the periods of stability when kernel deposition is strongest. Further experiments could also find the behaviour of dynamics in phase space $\alpha/\beta/\gamma/t_\theta$ for different neurological conditions. For instance, it is well-known that schizophrenia effects the

Summary

After examining our proposed model in depth, we have now compared to real human gaze data. This work has provided a dynamical explanation of the recently discovered significant regularity of saccades during a search task as a coupling between model dynamics and information through kernel deposition. We then made the novel prediction of gamma-scale oscillations in fixation dynamics, which was validated under the present data analysis. This component of fixational dynamics has rarely been studied as a coherent aspect of neural dynamics; we are the first to provide a modelled explanation of its neural origin. Finally, we presented the prediction of our model of unified fluctuation-driven superdiffusive motion across all spatial scales. By analysing fractional q–moments, we found that gaze trajectories are characterised by a single superdiffusive exponent across all scales, unifying the dynamics of fixations and of saccades under a single model. Finally, we suggest further experimental methods to test further quantitative predictions of the model designed to examine attention in the brain through the study of gaze dynamics.

Chapter 4: Functional Advantages of Visual Search Dynamics

Now that we have developed and validated a rigorous model of the dynamics of gaze across all spatiotemporal scales, we investigate what we can learn from the efficiency of the model to an informative search task. The first section of this chapter (Section 4.1) presents informational-theoretic methods which have been constructed to analyse efficiency during abstract search tasks. This method takes the framework introduced in Chapter 2, and using the information encoded in kernel dynamics, quantifies the information of a belief distribution as the distribution entropy and tracks its development through a search task. We take these metrics and compare the informational advantages of each dynamical pattern by comparing an abstract sampler operating in different points in parameter space $\alpha/\beta/\gamma/\tau$. The final section of the main body of this thesis (Section 4.2) examines further extensions of these dynamics beyond the case of a 2-dimensional visual search to high-dimensional or dynamical target landscapes, where control and exploration are hypothesised be more responsive and robust to shifts in a loss function.

4.1 Functional roles of dynamical sampling patterns

The dynamics studied in this thesis are characterised by three spatiotemporal signatures: theta-scale rhythmicity, gamma-scale local oscillations, and superdiffusive spatial trajectories. The framework of Chapter 2 constructs a biologically motivated model in which these signatures emerge from the underlying physical dynamics, and Chapter 3 found the same signatures in real human gaze trajectories, especially during realistic search tasks. Here we treat the model as a sampling theory and ask what each signature contributes to an abstract sampling process within neural computation. Each signature corresponds to a model parameter: the momentum coupling β generates local oscillations, the tail exponent α controls superdiffusion, and the timing of information updates sets the rhythmic organisation of sampling. To quantify their roles, we define information-theoretic metrics of search efficiency and describe how each signature affects them.

Information gain

The central quantity in the efficiency of a strategic visual search is the information gain, $\mathrm{IG}(t)$. To quantify the information gathered across an abstract search landscape, we draw on Bayesian inference and sampling methods. In particular, the link between the explored dynamic potential and a target-belief distribution (Equation (1.3)) gives a natural definition of the information gained during a search. To consider both the contributions from the cross-entropy of the belief distributions ($H(\pi_t, \pi_0)$), as well as the certainty in the entropy of the posterior belief ($H(\pi_t)$), we find usein the Kullback–Leibler (KL) divergence $D_{\mathrm{KL}}(\pi_t \parallel \pi_0)$, which has been used elsewhere in machine learning to define information gain

$$\begin{aligned}
\mathrm{IG}(t) &= H(\pi_t, \pi_0) - H(\pi_t) \\
&= -\int \pi_t(\mathbf{x}) \log \pi_0(\mathbf{x})\, d\mathbf{x} + \int \pi_t(\mathbf{x}) \log \pi_t(\mathbf{x})\, d\mathbf{x} \\
&= \int \pi_t(\mathbf{x}) \log \left(\frac{\pi_t(\mathbf{x})}{\pi_0(\mathbf{x})} \right) d\mathbf{x} \\
&\equiv D_{\mathrm{KL}}(\pi_t \parallel \pi_0). \qquad (4.1)
\end{aligned}$$

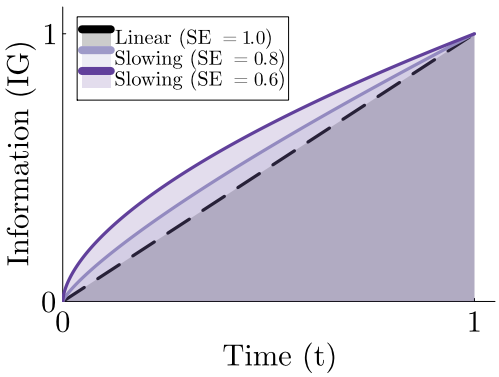


Figure 4.1: Schematic of dimensionless information gain. Time and IG are normalised to $[0, 1]$. The shaded area A gives $\mathrm{SE} = 1/A - 1$; the dashed line is $\mathrm{SE} = 1$.

The KL divergence is non-negative and asymmetric. In an abstract sampling model, non-negativity means that any update away from π_0 counts as accumulated evidence, though not necessarily as efficient search. A weak kernel, a slow information timescale τ, or deposition in low-likelihood areas alters the landscape only marginally and so produces slow growth in IG. The curve $\mathrm{IG}(t)$ therefore summarises the global information acquired as the observer's belief landscape evolves from π_0 to π_t, and the metrics in the following subsections capture different aspects of it.

4.1.1 Results

Metrics and comparison conditions

We collected metrics across an ensemble of searches, with 250 trials for each parameter set across the swept range. We considered only a single sweep across the conditions otherwise stated in Section 2.1: an α–β sweep, with α from 1.1 to 2.0 in increments of 0.05 and β from 0 to 1200 in increments of 100.

We use three metrics that capture complementary aspects of performance: the speed of completion, the speed at which information is gathered, and the temporal profile of information gain. The first passage time (FPT) is the time from trial onset to the first fixation on a target, perhaps the clearest measure of success, in seconds. The information velocity $\mathrm{IV} = \langle \mathrm{dIG}/\mathrm{d}t \rangle$ is the mean rate of information gain, in inverse seconds. The shape exponent SE describes the profile of the IG curve: we normalise time and IG to $[0, 1]$, compute the area A under the normalised curve, and define $\mathrm{SE} = 1/A - 1$. This is the exponent of the power law $\mathrm{IG} \propto t^{\mathrm{SE}}$ that has the same area as the observed monotonic curve. Thus $\mathrm{SE} = 1$ corresponds to accumulation at a constant rate, $\mathrm{SE} < 1$ to decelerating information gain, and $\mathrm{SE} > 1$ to accelerating information gain. Lower FPT and higher IV and SE indicate better performance, the latter meaning that the rate of information gain is better sustained rather than saturating early.

As an analytical reference for optimal FPT, we consider a searcher that fixates points uniformly at random without replacement, one per theta interval. For K targets among N points, the expected number of fixations up to and including the first target is $(N+1)/(K+1)$[1] and so, for $\tau = 0.2\,\mathrm{ms}$,

$$\mathbb{E}[\mathrm{FPT}_0] = t_\theta \, \frac{N+1}{K+1} = 6.0919\,\mathrm{s}, \tag{4.2}$$

where $t_\theta = 142.22\,\mathrm{ms}$ is the mean saccade interval for $\tau = 0.2$ ms.

Table 4.1: Correlations between key dynamical parameters and information metrics. Correlations are computed between each parameter and the median or mean of each metric across $n = 250$ trials. 95% bootstrap confidence intervals are shown in brackets below each estimate.

	Median			Mean		
Parameter	ρ_{FPT}	ρ_{IV}	ρ_{SE}	ρ_{FPT}	ρ_{IV}	ρ_{SE}
Momentum β	−0.300	0.641	0.508	−0.475	0.767	0.812
	[−0.633, 0.091]	[0.287, 0.891]	[0.148, 0.797]	[−0.720, −0.176]	[0.445, 0.951]	[0.577, 0.945]
Superdiffusion α	0.554	−0.748	−0.761	0.790	−0.951	−0.880
	[0.262, 0.712]	[−0.898, −0.544]	[−0.895, −0.607]	[0.640, 0.898]	[−0.981, −0.911]	[−0.953, −0.775]

Spatiotemporal strategies and their impact on search efficiency

Table 4.1 reports the correlation between each parameter and each metric across the tested range. Decreasing values of FPT represent a faster average time to success, increasing IV represents faster information gathering, and SE closer to 1 represents a search which doesn't decelerate, whereas SE

[1] This is the expected number of failures before the first success in sampling without replacement from a hypergeometric population, $(N - K)/(K + 1)$, plus the final successful fixation. See for instance [184].

closer to 0 represents a learning curve which flattens over a trial, getting trapped exploring low probability areas. We show results both for the median and mean of these metrics, to ensure a full view of the typical statistics, and ensure results aren't sensitive to outlier influences.

Momentum and decorrelation (β)

The coupling β sets how strongly the momentum variable influences the dynamics of the position of the searcher. Because momentum will tend to accumulates and respond to drift, it gives the trajectory inertia: rather than relaxing directly into the nearest basin of the potential, the searcher dynamically explores it. These same momentum-based mechanisms which decorrelate future samples drive the efficiency of successful sampling algorithms such as Hamiltonian Monte Carlo [185] and second-order Langevin sampling [186], where momentum reduces the correlation between successive samples so that a fixed number of samples covers more of the distribution. In a fixation, such decorrelation is effective in responding to a depletion of information locally. From a biological perspective, it has been shown that such rapid fixational dynamics and decorrelation can produce the high resolution of the visual system, especially in high contrast or fine-grained detail in visual landscapes [70].

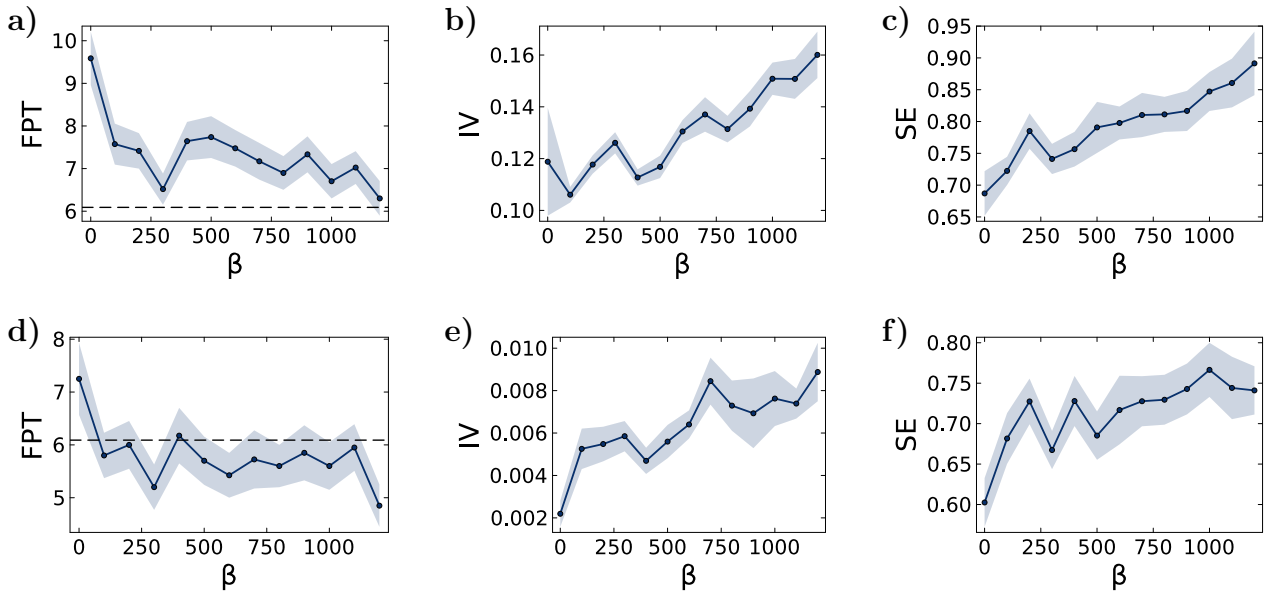


Figure 4.2: Trends in the three information metrics with local momentum driven oscillations β. Top row (**a**–**c**): mean; bottom row (**d**–**f**): median. Columns show FPT, IV and SE, left to right. The dashed line in **a)** and **d)** represent the expected FPT from an ideal independent sampler, derived in Equation (4.2).

The results are consistent with this interpretation (Figure 4.2). The local measure, information velocity, correlates strongly with β ($\rho_{\mathrm{IV}} = 0.64$ for the median and 0.77 for the mean), as does the shape exponent ($\rho_{\mathrm{SE}} = 0.51$ and 0.81), indicating that gain is sustained rather than saturating early through repeated sampling of the same region. FPT decreases with β, but this is the weakest effect: only the mean-based correlation (-0.48) has a confidence interval excluding zero. This is expected from the interpretation given above of local sampling having a limited effect on global exploration. It should be said, that the existence of oscillations does seem critical however, as will be presented later in the form of a periodic boundary, which has a pronounced effect on the efficiency of dynamics. Momentum therefore has the clearest benefit for the efficiency of information gathering, and a more modest benefit for the time to reach the target. The stronger mean-based correlations suggest that its main effect is to remove slow outlier searches.

Superdiffusion and global access (α)

The tail exponent α controls how heavy-tailed the jumps of the searcher are. At $\alpha = 2$ the noise is Gaussian and the searcher is Brownian, so reaching a distant region requires traversing everything in

between. For smaller α, occasional long jumps give direct access to any part of the landscape. For sparse unknown targets this strategy yields two benefits: the searcher is not confined to the region it has already sampled, and it can leave a region rapidly once remaining information has been depleted rather than diffusing slowly out of it. We therefore expect lower α to shorten FPT and to sustain information gain.

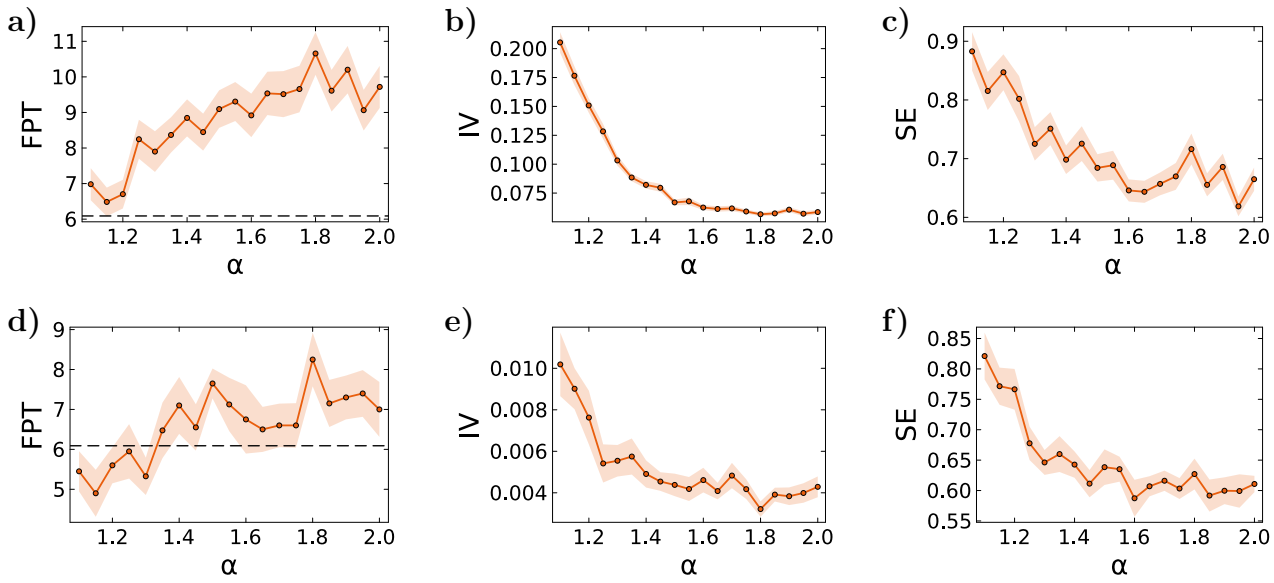


Figure 4.3: Trends in the three information metrics with superdiffusion α. Top row (**a**–**c**): mean; bottom row (**d**–**f**): median. Columns show FPT, IV and SE, left to right. The dashed line in **a)** and **d)** represent the expected FPT from an ideal independent sampler, derived in Equation (4.2).

All three metrics show this trend (Figure 4.3). FPT increases with α ($\rho_{\mathrm{FPT}} = 0.55$ for the median, 0.79 for the mean), while IV ($\rho_{\mathrm{IV}} = -0.75, -0.95$) and SE (SE $= -0.76, -0.88$) decrease. The correlation with IV is the strongest in Table 4.1, which suggests that long jumps also raise the local rate of information gain, plausibly because they explore regions more rapidly, and discover high probability modes. We therefore recover, in the presence of our additional modelled dynamics, the result that a sparse search is best served by a superdiffusive trajectory. The trend is approximately monotonic over the tested range, so this sweep does not by itself identify an interior optimum.

Joint nested oscillations and superdiffusion

The effects of β and α reported above are not independent. Figure 4.4 shows the joint dependence of the two metrics of information gathering, SE and IV, on both parameters. At high α, where search fluctuations are close to Brownian, performance is sensitively dependant on β. Both metrics are maximised along a narrow curve in the (α, β) plane, which marks where oscillations begin in the trajectory, as defined by Equation (2.11), and fall away for β larger or smaller. As α is lowered, this ridge broadens and is progressively smeared, until at the lowest values of α the metrics vary only weakly with β. This is a qualitative observation from a single sweep, but it is consistent across both metrics.

A likely interpretation is that the two mechanisms contribute at different scales. Momentum improves how a region is sampled locally, but a sampler that relies on it alone must be tuned so that its oscillations fit the scale of the landscape, much as the trajectory length

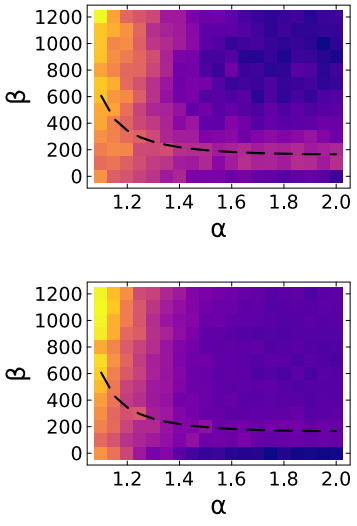


Figure 4.4: Joint effect of α and β...on search metrics...?. Median SE (top) and IV (bottom). Dashed line represents parameter bound for oscillations.

of Hamiltonian Monte Carlo must be tuned. Heavy-tailed jumps instead provide global access whatever the local dynamics, so that a poorly chosen β costs less when the sampler can leave an unproductive region directly. Lévy superdiffusion therefore makes the search robust to the choice of momentum regime, rather than requiring that regime to be tuned. The coupling may be computationally beneficial for this reason. Superdiffusion supplies global access, momentum supplies local refinement, and together they give a search that performs well over a wide range of parameters instead of at a single finely tuned point. This is the property sought in the minimally tuned sampler outlined in Section 4.2.2.

Summary

Taken together, the dynamical signatures—superdiffusion and oscillations—studied in this chapter play complementary roles in an efficient search. Momentum coupling β decorrelates successive samples, raising information velocity and sustaining the rate of information gain. Superdiffusion at low α gives global access to the landscape, improving all three metrics over the tested range, most clearly the speed of information gain. Rhythmic organisation regularises the timing of samples, complementing these spatial mechanisms.

4.2 Further extensions of studied dynamics

The dynamics studied in this thesis were introduced to explain gaze, but each of their components is a general mechanism for exploring a sparse and unknown landscape. Nonlocal superdiffusive jumps give access to distant regions, momentum-driven local oscillations decorrelate samples within a region, a history-dependent kernel records the information gathered and discourages revisiting, and theta-rhythmic transitions organise when the sampler moves between regions of interest. Nothing in their construction is specific to vision, and the model can be extended to a state space of any dimension. In this section we outline how these components might carry over to two settings in machine learning and artificial intelligence: sampling from difficult probability distributions, and pursuing goals that change in time. These are proposals for future work and are not tested here.

4.2.1 Dynamical goals

In the search tasks considered in this study, the goal is fixed: the landscape $V_0(\mathbf{x})$ and the target regions are set before the search begins, and only the observer's belief about it changes. Generally—including in machine learning—goals will be changing in time, as targets move, rewards shift, or agents switch tasks. A searcher must then remember where it has looked, to avoid wasting samples, and account for fact that old information may be less useful than new information. This motivates time-dependent landscapes $V_0(\mathbf{x}, t)$ which are intrinsically dynamical, beyond changing beliefs about the landscape. The information kernel already provides memory, and so including a mechanism of forgetting is simple; let the kernel effect decay back down to zero over time. Writing the potential as the current goal plus a kernel term recording the search history, $V_t(\mathbf{x}) = V_0(\mathbf{x}, t) + K_t(\mathbf{x})$, we let the kernel effect decay at a rate λ:

$$\partial_t K_t(\mathbf{x}) = \frac{1}{\tau} k(\mathbf{x} - \mathbf{x}_t) - \lambda K_t(\mathbf{x}), \qquad K_0(\mathbf{x}) = 0. \tag{4.3}$$

The case $\lambda = 0$ recovers the present model. For $\lambda > 0$, kernel deposition decays exponentially with memory rate $1/\lambda$ in units of time. This introduces two parameters into kernel dynamics, one (τ) which defines the theta-rhythmic rate of information gathering, and the second (λ) which defines the decay rate of this information in dynamical tasks. This model can be used to make rigorous predictions about the effect of working memory on theta-scales in behavioural experiments involving dynamical goals, such as the classic experiment requiring tracking a moving pendulum [187].

4.2.2 Superdiffusive probabilistic sampling

A central task in Bayesian inference and generative modelling is to draw samples from a distribution $\pi(\mathbf{x}) \propto \exp[-V(\mathbf{x})]$ that may be multimodal, with well-separated modes, or heavy-tailed [140, 185].

FNS was developed in order to solve this problem in cortical activity dynamics, as exploring well-separated heavy-tailed distributions are the central dynamics of neural sampling (see Section 1.3.1 [33]). From the field of abstract probabilistic sampling, standard gradient-based Markov chain Monte Carlo (MCMC) methods are driven by Brownian noise and therefore explore locally. A well-understood short-coming of these methods is that the time to escape a mode grows exponentially with the height of the surrounding barrier, so samplers can remain trapped in a single mode for very long periods if not tuned correctly [140]. Furthermore, heavy-tailed distributions give small gradients, which slow the exploration, and often fail to constrain standard MCMC algorithms [188].

Each component of the model addresses part of this problem, and this chapter has shown that each component efficiently provides an efficient strategy in searching through a complex multimodal landscape. First, driving the dynamics with α–stable Lévy noise ($\alpha < 2$) replaces local normally diffusive exploration with global superdiffusion, which increases the access to far displaced modes of interest. For the Lévy flight of this model, escape from a mode is governed mainly by the distance to the next mode rather than by the height of the barrier between them [160, 188]. Jumps between separated modes occur at a rate that does not decay exponentially with the barrier, but polynomially with the separation [160, 188]. Second, the momentum coupling produces fast local oscillations within a mode. As in Hamiltonian Monte Carlo, these decorrelate nearby samples, so that a fixed number of samples covers more of the local structure of π [160, 185]. Third, the dynamical-information kernel of Section 2.2 adds a history-dependent term to the potential, which penalises revisited regions. For extended sampling, this kernel can be allowed to relax down after the mode has been explored. This kernel deposition closely resembles metadynamics [189], in which deposited bias fills explored minima and pushes the sampler towards new ones. Finally, the regularity of a sampler with nested timescales has not been presented in the literature so-far, and this would represent a novel addition, allowing a decoupling of local sampling and the global exploration which modulates it, rather than through a hand-tuned mixture of local and global proposals, or a single local sampler which fails to explore multiple modes.

All of these dynamics can naturally produce an efficient, minimally tuned algorithm, only requiring a spatial scale defined by the covariance of an explored target, from which to set noise strength, and ensure local harmonic exploration. Developing a superdiffusive sampler is a active area of research in modern MCMC literature [142, 160, 188], and constructing a model based on the present equations of motion (Equation (2.1)) is a valuable addition to this literature. A full study in this direction would need to consider the following two components of the model: firstly, the information kernel adjusts the sampled distribution, and this would need be accounted for through, for instance, reweighting as in metadynamics [148, 149, 189], and secondly, calculation of the full fractional derivative in the gradient drift term may be computationally prohibitive. There exist creative methods to account for both considerations, but such a study is left to further work.

Chapter 5: Discussion

In this thesis, we have developed a unified model of the rich multiscale dynamics of gaze during active visual search. Previous studies have largely addressed two distinct regimes of gaze behaviour: the relatively slow, large-scale movements associated with saccades, and the rapid, fine-scale dynamics occurring during fixations. Here, we bring these regimes together within a single physics-inspired theoretical framework. Moreover, previous approaches have often emphasised either the biological mechanisms underlying visual search or the development of generative models capable of reproducing empirical eye-tracking data. Our approach seeks to bridge these perspectives by deriving a predictive model of gaze from the cortical dynamics described by the theory of fractional neural sampling (FNS) [11, 33], and developing methods to evaluate its predictions against human eye-tracking data.

From the perspective of statistical physics, the dynamics examined in this thesis provide an experimentally accessible example of a complex, nonequilibrium system exhibiting anomalous transport, nested oscillations, and history-dependent behaviour. Three principal features are of particular importance. First, visual dynamics are shown to be superdiffusive, and moreover display robust monoscaling, governed by a single stability index α. Such anomalous diffusion at every scale is of interest when studying how basic computational dynamics may be constructed from complex dynamical systems at criticality. We characterised the short-time diffusive scaling of eye-tracking data through fractional absolute moments (Section 3.3), and compared these to the monoscaling superdiffusion of Lévy dynamics (Section 2.4). Such rich multiscale self-similarity provides a physical explanation as the result of control from a neuronal network at criticality with diverging correlation length, resulting in nonlocal spatial access to an informative landscape [190–194].

Second, the model presents and explains the unique pattern of nested oscillations: the slow theta-scale rhythmic transitions between regions of interest, which modulate fast gamma-scale local sampling dynamics, arising from the broad patterns of neural computation, and the confinement of a sampling process. Nested timescales are a recurrent feature of coherent neural dynamics, especially during communication of information between areas of the brain [9, 59, 84–87]. The present model carries this framework over to gaze dynamics, as they emerge from the same neural dynamics which control the process of active vision.

Third, the dynamical-informative kernel introduced in this work describes how visual dynamics are influenced by the history of the gaze trajectory. In particular the informational efficiency of gaze trajectories and the phenomenon of inhibition of return (IoR) suggest a strong dependance on the informative history of a search. Our formulation naturally encodes this information through a belief landscape which evolves from a prior saliency π_0 to a posterior belief after exploration π_t. The cumulative information gathered is then well defined through the Kullback–Leibler (KL–) divergence. Such a well-motivated physics-inspired development of interpretable memory-effects, heavy-tailed noise, and nonlocality is important in the development and progress of physics as an interdisciplinary field.

This work advances the study of the brain in neuroscience, as our model turns gaze into a source of interpretable parameters modulating neural attention, a very important topic of study when understanding decision-making in the brain. The active parameter space here defined through the stability index α, memory–momentum strength β, noise scale γ, and informative timescale t_θ: each parameter has a well-understood and motivated interpretation in neural dynamics. Because gaze aggregates the activity of many brain regions, designing experimental paradigms to fit parameters and summarise collective properties of the circuits that control and process visual input is very impactful to take non-invasive measurement of the brain.

Experiments have been suggested through this work to apply this framework in understanding brain function. An experimental protocol tracking high-frequency eye movement alongside coarse-grained saccade positions during a standardised search task without head restraint, combined with head ac-

celeration measurements, would resolve gamma-scale fixational dynamics in naturalistic, head-free viewing and clarify the role of OMT during search. During such an experiment, coherence of the high-frequency sampling during fixations can be examined rigorously, as well as testing the saliency-frequency coupling and mapping the paramter-space dynamics as described in Section 3.4. Such an experiment is simple to design using new low-cost and non-invasive techniques such as described in [58] and the data extracted is tractable to analyse across the scales of interest here, to test the predictions of FNS dynamics here presented.

The same sampling process applies to machine learning, in particular to sampling accurately from a given posterior distribution $\pi_{\text{post}}(\boldsymbol{\theta})$ over a set of parameters $\boldsymbol{\theta}$, a problem often handled through the use Markov chain Monte Carlo (MCMC) sampling. Standard MCMC methods are based on sampling a chain of points in parameter space, and proposing local moves through a probability landscape $V(\boldsymbol{\theta}) = -\ln \pi_{\text{post}}(\boldsymbol{\theta})$, often incorporating momentum-coupled Langevin dynamics through parameter space as introduced in Section 1.2. Such dynamics often are unable to robustly explore and mix between well-separated modes, and designing an algorithm to robustly explore a wide range of unseen heavy-tailed or multimodal distributions efficiently without fine-tuning of algorithmic hyperparameters remains a long-standing and active topic in machine learning [139, 140, 195]. The superdiffusive sampling of this thesis address this issue directly, as we have found that in an unseen search task, superdiffusive jumps will efficiently gather informative samples. The momentum-driven local oscillations have also been shown to improve sensitivity to sampled information and efficient sample mixing, as found in similar momentum-driven dynamics in Hamiltonian Monte Carlo [185, 196] and second-order Langevin sampling [186, 196]. Further, the adaptive kernel resembles well-tempered metadynamics [189], in which deposited bias discourages revisiting explored regions, robustly improving ergodicity without reliance on parameter tuning on the unknown landscape $V(\boldsymbol{\theta})$. The link to free-energy minimisation and active inference [42, 149, 189, 197, 198] is important when formalising this link, since both theories consider search as a minimisation of uncertainty about a landscape.

Our results emphasise that coupling oscillations across scales is itself important for an abstract computational sampler: rapid local oscillations explore each mode, while slower rhythmic transitions decide when to move on. Superdiffusive dynamics in generalised computational systems have recently attracted considerable interest, both from theoretical studies of such dynamics emerging from informational sampling [103, 122, 199, 200] as well as in designing algorithms in MCMC sampling of abstract distribution landscapes [140, 142, 160, 188, 201]. The same principle of heavy-tailed nonlocality in computation has also influenced more general AI frameworks, notably in work from our group on Fractional Neural Attention (FNA) [202]. FNA is designed to token interactions through Lévy diffusion, as governed by the fractional Laplacian, so that short- and long-range dependencies arise from a single operator rather than hand-designed local–global patterns or fine-tuned scales of influence. In the reported experiments it matches or exceeds Transformer baselines on text classification, image classification, and machine translation, in some cases with a single layer, and its attention networks show larger spectral gaps and shorter path lengths, consistent with faster information mixing. That the same mechanism describes both gaze and attention in artificial networks supports the view that multiscale, nonlocal sampling is a general computational strategy.

In conclusion, this thesis demonstrates that the apparently disparate dynamics of human gaze can be described within a unified physical framework comprising a small number of interpretable mechanisms: nonlocal Lévy motion, nested oscillatory dynamics, and an adaptive information-dependent sampling landscape. By connecting these mechanisms to the theory of fractional neural sampling, the model offers a quantitative account of how complex neural dynamics may give rise to the coordinated exploration of visual information across spatial and temporal scales. As a readily observable behavioural signal, gaze provides a valuable window into aspects of neural attention and decision-making. The framework developed here establishes testable predictions for future experiments, provides new opportunities for the non-invasive characterisation of neural function, and suggests broader applications of multiscale stochastic dynamics to probabilistic computation and artificial intelligence.

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Appendix A: Prior models of visual search

Model	Core domain	Primary aim	Key features
Ideal Bayesian searcher (IBS) Najemnik and Geisler [37], see also [35, 36, 152]	Information- and decision theory	Establish an optimally informative process for saccade localisation based on biologically realistic information foraging of the eye.	Exact Bayesian belief updating, incorporating biologically realistic retinal resolution gradients, spatial noise, and global entropy reduction. Considers discrete saccades rather than continuous time.
Least action / active inference Zanca and Gori [42]	Machine learning	Generating scanpaths (trajectories) through physics-inspired variational informational free-energy minimisation.	Generative deterministic model of scanpaths. Sophisticated account of natural image saliency. Does not consider fixational dynamics, does not aim to be biologically realistic.
Self-avoiding random walk Engbert et al. [7], see also [77]	Statistical physics and oculomotor control	Develop an accurate account of low-frequency ocular drifts during free-viewing fixations.	Lattice-based stochastic motion with self-avoidance constraints; 2D quadratic potential. Does not capture a high-frequency tremor or saccades.
DeepGaze Abdelrahman et al. [203], see also [47, 48, 204, 205]	Deep learning and computer vision	Construct a neural network to predict human scanpaths on natural images with high empirical accuracy.	Modular deep neural network trained on eye-tracking data; models internal state, memory, and information. Not designed as a biologically realistic model; considers discrete saccade locations.
Itti–Koch–Niebur model Itti et al. [74]	Biologically inspired vision	Simulate bottom-up, stimulus-driven visual attention based on early primate visual pathways.	Multi-scale feature extraction (colour, intensity, orientation); winner-take-all selection with inhibition of return (IOR).
Feature-integration theory Gayet and Aydemir [76]	Cognitive psychology	Explain experimental differences in performance during sparse- and dense-search tasks.	Pre-attentive direction of attention based on saliency theory. Biologically motivated, but not generative or explanatory for eye movement measurement.
Guided search Wolfe [75] and Wolfe et al. [206]	Cognitive psychology	Account for the continuum of multi-scale search efficiency by combining bottom-up and top-down guidance of attention.	Parallel feature maps weighted by stimulus salience and target template to form an activation map; serial deployment of attention in decreasing activation order.

Table A.1: Review of a selection of influential models of visual search and gaze dynamics. Comparison across core theoretical domains, primary objectives, and defining operational features. Models span a spectrum from biologically grounded frameworks prioritising physiological mechanisms (e.g., Feature-Integration Theory, Itti–Koch–Niebur) to normative and computational architectures optimised for empirical trajectory prediction on complex natural scenes (e.g., Ideal Bayesian Searcher, DeepGaze). Important features which share analogy in the model presented in this thesis include: an information kernel (in discrete time) from the approach in Najemnik and Geisler [37], the second-order (noiseless) dynamical system coupled to a rich account of saliency gradients presented in Zanca and Gori [42], and the description of the rich non-tremor fixational random motion in the presented in Engbert et al. [7].

Appendix B: Code availability

Repository	Function
VisualFNS	Producing simulations, visualisations, and raw data of gaze trajectories through numerically integrating the system Equation (2.1) with kernel adaptation under Equation (2.5). Important techniques in solving this system were the use of the Euler–Maruyama numerical scheme for SDE integration and the Fourier-space spectral solution of kernel updates.
VisualFNS_Analysis	Analysing trajectory data which was output from VisualFNS. Involved producing power spectra, marking intervals between saccade-like transitions, and analysing superdiffusive scaling.
VisualData_Analysis	Loading in the data provided from Näher et al. [61], and analysing it in the same way as in VisualFNS_Analysis. Involved producing power spectra, marking intervals between saccade-like transitions, and analysing superdiffusive scaling. Originally forked from FractionalSaccades.jl.
FractionalNeuralSampling.jl	The project which provides the computational basis to formalise the FNS equation and its variants in syntax solvable through DifferentialEquations.jl.

Table B.1: A list of repositories containing the code used in this thesis. All code was written in Julia, apart from some helper scripts written in Python.

Appendix C: Eye-tracking supplements

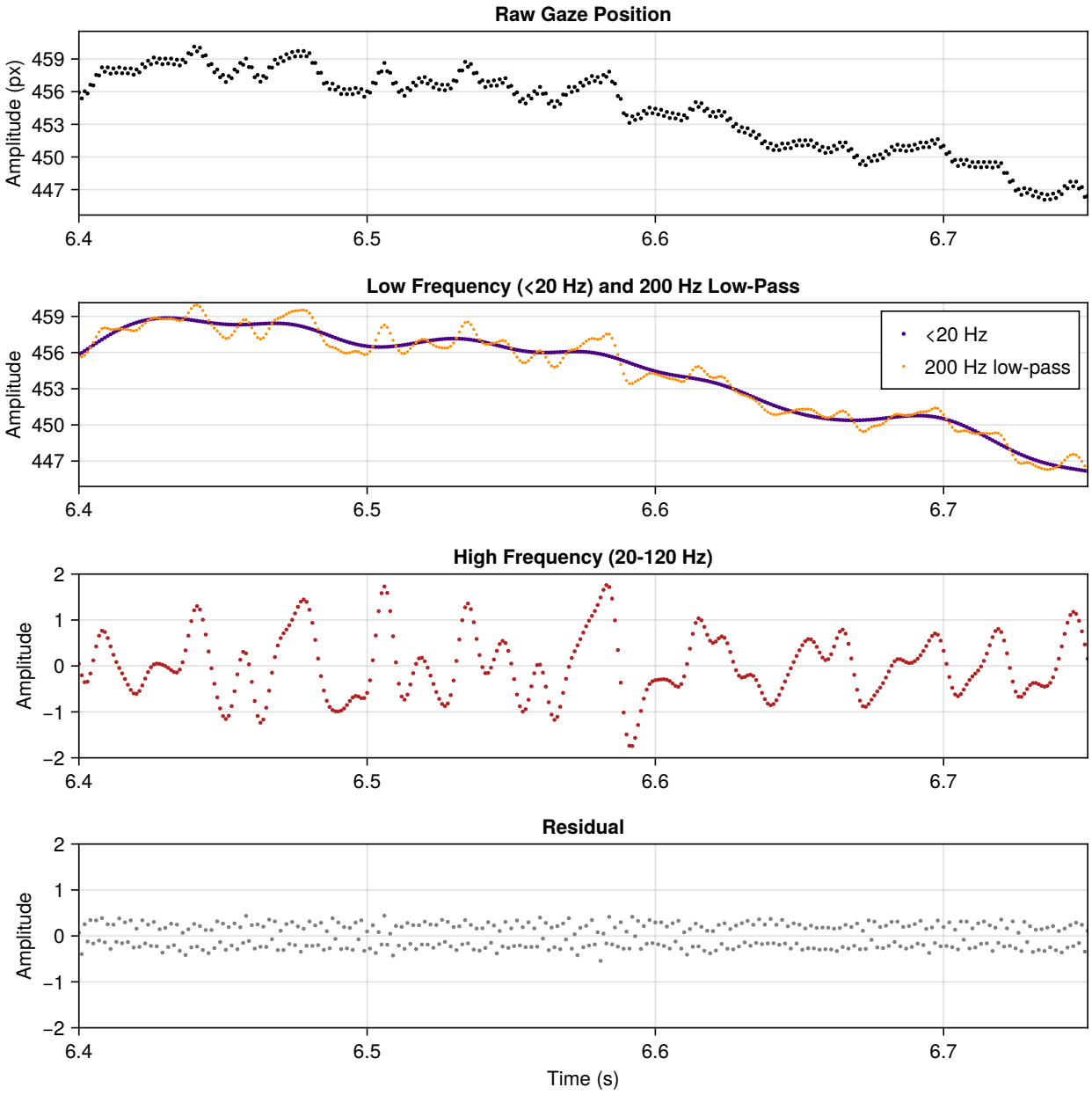


Figure C.1: The Fourier decomposition of a typical fixation.